Dundalk Institute of Technology

An Exploratory Analysis of Participation Patterns in European Citizens Initiatives using Country-Level Socio-Economic Variables

Thesis August 2026

Student Name: Matthew Riddell

Student ID: D00245674

Programme: MSc in Data Analytics

Module: Dissertation

Supervisor: Dr. Catherine Mulwa

Submission Date: August 30, 2026

Contents

Abstract

The European Citizens' Initiative (ECI) provides EU citizens with a mechanism to propose changes to European policy. However, participation varies significantly between countries and Initiatives. This dissertation investigates whether country-level socio-economic indicators can explain differences in ECI participation. ECI signature data was collected through web scraping and combined with country-level socio-economic data. Following data preparation and EDA, linear regression, logarithmic regression, Principal Component Analysis (PCA), Random Forest regression and cluster analysis were created and validated. Participation was examined using both raw signature counts and signatures per capita. A separate dataset containing only successful Initiatives was also analysed. The results showed that socio-economic indicators had explanatory power for raw signature counts with population being the dominant factor. Explanatory power was substantially lower when participation was measured using signatures per capita. Regression and Random Forest models generally produced weak predictive performance across the complete dataset. Some improvement was observed when analysing only successful Initiatives, particularly using logarithmic PCA regression, although the smaller dataset limited the reliability of these findings and Random Forest models showed evidence of overfitting. The findings suggest that country-level socio-economic characteristics alone cannot adequately explain ECI participation. The variation between Initiatives indicates that factors such as campaign organisation, mobilisation, political context and issue factors may play a more important role. The dissertation demonstrates the iterative data analytics process to investigate patterns of political participation and provides a foundation for future research incorporating Initiative level and campaign related factors.

Keywords: European Citizens' Initiative, Data Analytics, CRISP-DM.

1 Introduction

1.1 Background

Political participation is a key component of democratic systems, allowing citizens to express their preferences, influence political movements and communicate demands to decision makers. Online petitions have grown as a popular form of political participation in recent years, providing citizens with a legitimate means through which support for a political issue can be expressed and heard.

At the European Union level, the European Citizens' Initiative (ECI) provides a formal mechanism through which citizens can seek to influence the policy and legislature of the European Commission. An ECI allows EU citizens to request that the European Commission consider proposing legislation on a particular issue. To progress through the ECI process, an initiative must obtain at least one million valid signatures from EU citizens and pass the minimum thresholds of signatures in at least 7 European Union member states (European Commission, n.d.; Europa.eu, 2024).

Although individual initiatives address issues relevant across the European Union, participation does not necessarily occur evenly between member states. Some countries may contribute much higher proportions of signatures than others, while the same country may display very different levels of participation depending on the initiative being considered.

This variation creates a new opportunity for data analytics to investigate whether or not measurable country-level indicators can help explain differences in participation. Economic factors such as GDP and unemployment may potentially influence participation, while demographic based factors like population and population density may also affect the number of citizens available to participate. Other factors such as education, governance, personal freedom, social capital and media literacy may also provide insights and potential explanations for variance in participation.

However, the issue is rather complicated, by the fact that the absolute number of signatures collected from a country is strongly influenced by it's population. A larger country like Germany would have a much higher potential population of participants compared to a much smaller country like Luxembourg. As a result, comparing countries using raw signature counts alone may capture mostly differences in population size rather than differences in the willingness of citizens to participate in ECIs.

For this reason, this project considers both raw signature counts and signatures per capita. Raw signature counts provides a measure of the absolute participation of a country to an initiative, while signatures per capita provides a measure of participation that has been normalised by population. This distinction allows for the analysis to investigate whether or not country-level indicators explain the overall signature counts or whether or not they provide evidence of differences in participation relative to population sizes.

The initial analysis considered variables including population, GDP, GDP per capita, population density, life expectancy, unemployment and tertiary level enrollment which were adapted from the World Data 2023 dataset (Elgiriyewithana, 2023) and analysed and used to create models to answer the interim research questions..

The project developed significantly beyond this initial stage. Exploratory analysis revealed that population had a moderate relationship with total signatures, while the relationships between many of the country-level variables and participation were considerably weaker. More importantly, participation rates varied greatly between individual initiatives. This was investigated statistically during the interim stage of the project using an analysis of variance (ANOVA), which provided strong evidence that participation differed between petitions.

This finding was important as it indicated that participation could not be treated as homogeneous across all ECIs. Differences between initiatives could potentially reflect petition specific characteristics, such as the issue being addressed, the level of public interest or the ability of an initiative to mobilise supporters (Weisskircher, 2019; Hale, Margetts, & Yasseri, 2013).

The final stage of the project therefore followed an iterative approach. Additional ECI data was incorporated, expanding the dataset from 7 to 27 petitions. Additional country-level indicators were also introduced, including the Media Literacy Index 2023 and variables from the Global Development and Prosperity Index 2023 (Taşaltǐ, 2023; Lessenski, 2023).

The final project therefore, represents an iterative data analytics investigation. The results of the earlier stages informed the subsequent stages and the Interim report acting as an important development point in the overall process, informed a complete reiteration through all stages of the CRISP-DM life-cycle (Data Science PM, 2024).

1.2 Research Problem

The main problem investigated is whether or not country-level socio-economic indicators can explain differences in participation in European Citizens' Initiatives.

The initial problem is that raw signature counts are not directly comparable between countries because countries have different population sizes. A country such as Germany with a population of roughly 80 million people will naturally have a far greater potential pool of participants than a country like Cyprus with around 1 million people. As a result, a high number of signatures doesn't always mean a higher level of participation among it's population.

To address this, a signatures per capita variable was created:

$$\text{Signatures Per Capita} = \frac{\text{Signatures}}{\text{Population}}$$

This variable provides an alternative representation of participation that accounts for differences in country population.

However, the analysis also identified a second problem: participation varies greatly between individual petitions. Therefore, a simple analysis of all combined initiatives may be obscured by petition specific differences.

The interim analysis found significant variation in petitions from the result of the ANOVA. This provided a basis for extending the analysis beyond just the original combined dataset and the inclusion of a secondary derived dataset containing only the successful petitions.

The final project as a result, considers participation at three levels.

First, it investigates whether socio-economic indicators can explain raw signature counts.

Second, it investigates whether socio-economic indicators can explain variation in signatures per capita across all initiatives.

Third, it investigates whether socio-economic indicators can explain variation in signatures per capita across only the successful initiatives.

This progression allows for the project to examine both the effect of population size and the potential effect of petition outcome on the relationship between country-level indicators and participation.

1.3 Research Questions

The following research questions were created to address the main research problems:

1.3.1 Research Question 1

Can socio-economic indicators explain raw signature counts?

This first question investigates whether differences in the number of signatures can be explained using socio-economic indicators.

This question is addressed via the creation and validation of three linear regression models:

(Linear Regression 1A, Linear Regression 1B, Linear Regression 1C)

1.3.2 Research Question 2

Can socio-economic indicators explain participation rates?

This second question investigates whether differences in participation rates can be explained using socio-economic indicators, when participation has been normalised by population.

This analysis uses the master dataset which uses both successful and unsuccessful petitions.

The following models were created and validated to answer this question:

(Linear Regression 2A, Linear Regression 2B, Linear Regression 2C, Log Linear Regression 2A, Log Linear Regression 2B, Log Linear Regression 2C, PCA Regression 01, PCA Log Regression 01, Random Forest Regression 01, Random Forest Regression 03 (Log))

1.3.3 Research Question 3

Can socio-economic indicators explain participation rates when the analysis is reduced to only successful Petitions?

This third question investigates whether differences in participation rates can be explained using socio-economic indicators, when participation has been normalised by population.

This analysis uses the successful dataset which restricts the analysis to only successful petitions.

The following models were created and validate for this analysis:

(Linear Regression 3A, Linear Regression 3B, Linear Regression 3C, Log Linear Regression 3A, Log Linear Regression 3B, Log Linear Regression 3C, PCA Regression 02, PCA Log Regression 02, Random Forest Regression 02, Random Forest Regression 04 (Log))

Structurally each model is identical to its question 2 counterpart, the only difference being the dataset selection. The comparison between question 2 and question 3 here is particularly important as it allows the effect of petition selection to be examined directly.

1.4 Research Objectives

To answer these questions, following objectives have been outlined for the project:

1. Collect and integrate ECI participation data from multiple European Citizens' Initiatives.
2. Integrate ECI data with country-level socio-economic datasets to create a combined dataset.
3. Clean and standardise the data, including country names, numerical variables and petition-level information.
4. Develop measures of participation, including raw signature counts and signatures per capita.
5. Explore relationships between participation and country-level indicators using descriptive statistics, correlation analysis and visualisation.

6. Investigate differences in participation between petitions, including the ANOVA analysis undertaken during the interim stage.

7. Develop linear regression models addressing raw signatures and signatures per capita.

8. Investigate the effect of logarithmic transformation on signatures per capita.

9. Apply PCA to investigate whether combinations of correlated socio-economic variables provide greater explanatory power than individual variables.

10. Apply Random Forest regression to investigate potential non-linear relationships and evaluate unseen predictive performance.

11. Apply cluster analysis to examine groups of countries with similar socio-economic characteristics.

12. Compare the results across the full dataset and successful petition dataset to determine whether petition outcome influences the observed relationships.

13. Evaluate the findings in the context of existing research and identify limitations and opportunities for future investigation.

1.5 Project Relevance

This project is relevant to Data Analytics as it applies a range of analytical techniques to a real world democratic participation problem using real datasets.

This project demonstrates how data from different sources can be collected, cleaned, integrated and transformed into a dataset that can be used for data analytical methods. It also demonstrates the importance of exploratory analysis before selecting and applying modelling methods.

Instead of just assuming that one modelling approach will provide a complete explanation of participation, the project applies multiple different modelling approaches from different angles. Regression models provide a statistical baseline for linear relationships, logarithmic transformations address the issue of skewness in participation, PCA reduces correlated variables into a smaller set of components, Random Forest provides a non-linear machine learning approach and cluster analysis provides an unsupervised examination of country-level variables.

This project is also relevant because the analysis concerns political participation at the EU level, an area in which understanding differences between countries can potentially provide insights and inform future decision making associated with citizen engagement.

1.6 Research Methodology

This project follows the CRISP-DM methodology, which provides a structured framework for conducting the dissertation as illustrated by the following diagram by (Jensen, 2012).

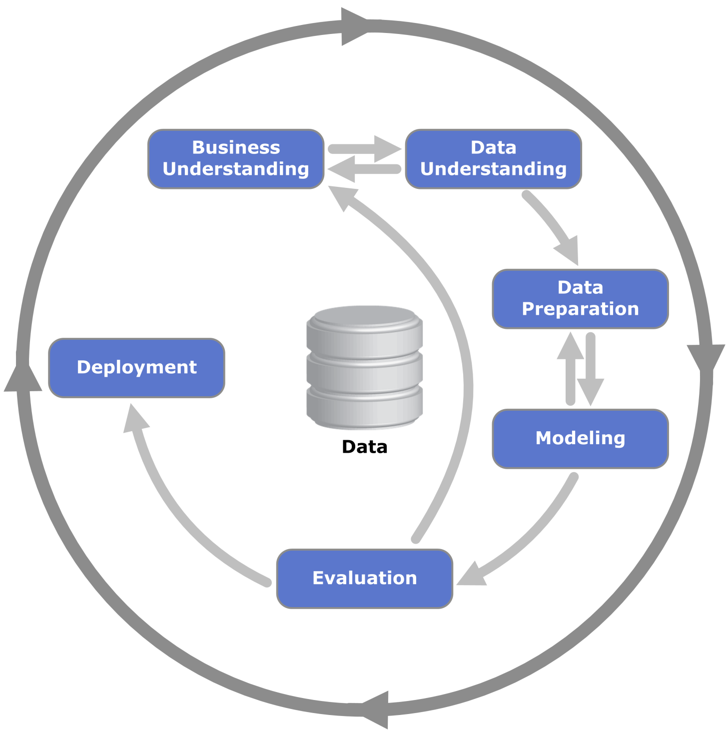

Figure 1: CRISP-DM process

Source: Jensen (2012), CC BY-SA 3.0.

The methodology was implemented through the following iterative stages:

1. **Web Scraping / Data Collection**
2. **Initial Data Exploration**
3. **Data Preparation and Integration**
4. **Exploratory Data Analysis**
5. **Feature Engineering and Data Modelling**
6. **Evaluation**

This structure was established during the interim stage of the project and was retained for the final analysis.

This process was treated as iterative rather than strictly sequential. Findings from the exploratory analysis influenced the selection of additional variables and modelling approaches. Similarly, findings from the initial modelling from the interim stage informed the use of transformations and an expansion into other modelling approaches such as PCA, Random Forest, and the creation of a derived successful petition only dataset.

The final project as a result, represents a complete reiteration through every stage of the life-cycle rather than a simple continuation directly where the interim stage left off, this included new scraping and data integration and preparation as well as new explorations and modelling. The interim report provided an initial understanding of the data, while the final stage expanded and refined the analysis based on the findings obtained.

1.7 Hypotheses

Raw Signature Counts

- $\mathbf{H_0}$: Socio-economic indicators do not explain a significant proportion of the variation in raw signature counts.
- $\mathbf{H_1}$: Socio-economic indicators indicators explain a limited portion of the variation in raw signature counts.

Signatures Per Capita (All Petitions)

- $\mathbf{H_0}$: Socio-economic indicators have no significant relationship with participation rates when participation is normalised by population.
- $\mathbf{H_1}$: Socio-economic indicators have at least a limited explanatory power when participation is normalised by population.

Signatures Per Capita (Successful Petitions)

- $\mathbf{H_0}$: Socio-economic indicators have no significant relationship with participation rates when participation is normalised by population.
- $\mathbf{H_1}$: Socio-economic indicators have at least a limited explanatory power when participation is normalised by population.

Structure of the Report

This report is structured as follows. Chapter 1 covers the introduction and goals of the project. Chapter 2 covers a review of relevant literature in relation to European Citizens' Initiatives, online petition participation and the use of data analytics within democratic engagement research. Chapter 3 describes the data collection process, initial exploratory data analysis, data cleaning and preparation, exploratory data analysis and modelling methods. Chapter 4 covers results, findings and evaluations of the models. Chapter 5 covers conclusions and future work.

2 Literature Review

2.1 The European Citizens' Initiative. Mobilization Strategies and Consequences

Weisskircher (2019) investigates the role of social movements and mobilisation strategies within the European Citizens' Initiatives (ECIs). The study investigates how campaign organisations attempt to mobilise support across European countries and it argues that successful initiatives tend to concentrate their mobilisation efforts in a small group of larger countries rather than attempting an evenly distributed EU-wide mobilisation effort. Through an in-depth analysis of the 2012 Initiative "Stop Vivisection" which successfully surpassed one million signatures, the research demonstrated that campaign organisation and mobilisation strategies are important factors that determine the success of an ECI. The study also highlighted that initiatives are able to generate political outcomes beyond legislative changes, including increased public awareness and institutional reforms (Weisskircher, 2019).

These findings are relevant to this project as they suggest that participation rates may very greatly between Initiatives due to campaign specific factors rather than only being determined by demographic or socio-economic factors. This supports the inclusion of multiple petitions within the dataset and it also provides justification for analysing differences between different Initiatives using statistical methods such as Analysis of Variance (ANOVA).

These findings are also relevant to distinction between raw signature counts and participation rates used in this dissertation. If mobilisation is concentrated withing larger EU member states, countries with larger populations may naturally produce greater numbers of signatures without necessarily having a greater proportion of their population participating. This provides justification for investigating both total signature counts and signatures per capita as separate measures of participation.

This issue became particularly relevant during the development of the dissertation. The ANOVA undertaken during the interim stage of the project identified substantial differences in participation between the Initiatives included in the dataset. This suggests that the individual Initiative itself may be an important source of variation and should be considered when interpreting relationships between country-level indicators and participation (Weisskircher, 2019).

2.2 Petition Growth and Success Rates on the UK No. 10 Downing Street Website

In this paper, Hale et al. (2013) analysed more than 8,000 online petitions submitted through the UK Government's No. 10 Downing Street petition platform. The authors used a large-scale data analytics approach to investigate petition growth patterns and identify factors associated with petition success. Their findings revealed that successful petitions generally experienced rapid growth during the early stages of their campaigns, with signatures collected during the first day being strongly associated with overall campaign performance. The study also showed that participation patterns were highly irregular and uneven, with a relatively small number of petitions obtaining subsequent support while the majority received limited engagement and failed to pass (Hale et al., 2013).

This research is highly relevant to the dissertation as it demonstrates how petition data can be analysed using statistical methods to examine patterns of citizen participation. Similar to this project, the authors utilised behavioural data generated through actual participation rather than relying exclusively on survey responses.

The study also highlights the importance of petition-specific factors when explaining participation outcomes. This is particularly relevant to the dissertation as the final dataset contains multiple ECIs concerning different political and social issues. Differences in the visibility, popularity and mobilisation of individual Initiatives may therefore contribute to variation in signature counts that cannot be explained by country-level socio-economic indicators.

The research also provides a methodological foundation for the analytical approach used in this dissertation. While Hale et al. (2013) primarily examined petition growth and success, this project extends the general quantitative approach by combining ECI signature data with external country-level socio-economic variables. Regression, dimensionality reduction, machine learning and clustering methods are subsequently used to investigate whether measurable country variables can explain differences in participation.

The significant variation observed in petition participation within the work of Hale et al. (2013) is also consistent with the ANOVA results obtained during the interim stage of the dissertation. This provides further support for treating differences between individual Initiatives as an important consideration rather than assuming that all petitions represent the same participation process (Hale et al., 2013).

2.3 Mobilizing Europe's citizens to take action on migration and climate change: behavioral evidence from 27 EU member states

Giebler, Giesecke, Humphreys, Hutter, and Klüver (2025), investigate citizen participation across all 27 EU member states using behavioural data collected through the European Parliament Election Study. The authors examine support for petitions relating to migration and climate change and investigate whether country-level factors can explain differences in participation between member states. Their research identified significant differences in support levels across countries and policy issues. However, the authors also reported that country-level factors demonstrated limited explanatory power when attempting to explain participation behaviour (Giebler et al., 2025).

The findings by Giebler et al. (2025) are particularly relevant to this dissertation as they closely correspond with the results obtained during the exploratory and modelling stages. The initial regression models developed during the Interim stage indicated weak relationships between participation rates and socio-economic indicators such as GDP per capita and unemployment.

The study suggests that participation behaviour may be influenced more strongly by issue-specific factors, political context and campaign factors than by country-level characteristics alone. This provides an important theoretical basis for interpreting the relatively weak explanatory performance observed in the final models.

This became particularly relevant as additional variables were introduced into the dissertation. The final dataset incorporated a broader range of country-level characteristics, including media literacy, governance, social capital, personal freedom and education. Despite this expansion, the regression models generally demonstrated limited explanatory power. This suggests that increasing the number of country level variables does not necessarily provide a substantially stronger explanation of participation.

The findings therefore provide an important comparison for the results of this dissertation. Weak predictive performance should not necessarily be interpreted as evidence that the analytical approach was inappropriate. Instead it may reflect the limitations of attempting to explain complex political participation behaviour using country-level socio-economic indicators alone (Giebler et al., 2025).

2.4 Identifying patterns and recommendations of and for sustainable open data initiatives: a benchmarking-driven analysis of open government data initiatives among European countries

Lnenicka et al. (2023) examine the development and sustainability of open government data initiatives across European countries. Using a benchmarking driven analysis of open data frameworks and indices, the authors identify patterns that influence the importance of open data as a resource for research and development, transparency and evidence based decision making (Lnenicka et al., 2023).

Although this study doesn't specifically examine political participation or petitions, it provides an important justification for the use of publicly available government data within this dissertation. The dissertation relies on publicly available data from the European Citizens' Initiative platform and combines this information with external socio-economic datasets.

The research therefore supports the use of open government data as a valuable resource for large scale analytical projects. It also demonstrates the importance of integrating data from multiple sources, which played a significant part in this dissertation.

The ECI data alone did not contain all of the variables needed to investigate the research questions. As a result, it was integrated with external country-level datasets to provide economic, demographic and societal indicators for each country. This process required country standardisation, conversion of numerical fields and removal of information that was irrelevant to the analysis.

The use of multiple open datasets therefore allowed the project to move beyond analysing petition signatures in isolation and investigate whether participation could be associated with broader country-level characteristics (Lnenicka et al., 2023).

2.5 Direct democracy and equality: context is the key

Geißel, Krämling, and Paulus (2023) explore the relationship between direct democratic mechanisms and equality across European countries. Using large scale statistical models, the authors investigate whether country-level variables can explain outcomes associated with direct democratic processes. Their findings suggest that country-level variables alone often fail to provide clear explanations for observed outcomes. The authors conclude that the contextual and case specific factors can play a much more significant role than broad socio-economic variables and they argue that statistical analysis should be complemented by an understanding of the social and political context in which participation occurs (Geißel et al., 2023).

This study is relevant to the dissertation as it demonstrates the limitations of relying solely on country-level variables to explain participation behaviour. The findings support the possibility that variables such as GDP, education and population may only partially explain participation patterns observed within ECIs.

This is relevant to the the creation of signatures per capita in the project. Population is naturally associated with the number of potential participants in a country, meaning that the larger countries can produce potentially far larger signature counts. Converting signatures into a per capita measure provides a more comparable measure of participation across countries, but does not eliminate the influence of petition specific or contextual factors.

The findings of Geißel et al. (2023) therefore provide an important theoretical explanation for why the predictive models developed within this dissertation may demonstrate relatively weak explanatory power despite considerable variation in participation between countries and Initiatives.

This study also provided much of the methodological framework for the methodologies employed within the project. The study makes extensive use of statistical methods and regression based modelling, which in a sense provided a foundation and starting point for the Interim stage of the project, which has since expanded beyond the use of solely linear regression models but also incorporated logarithmic transformations, dimensionality reduction, random forest regressions and cluster analysis in order to provide a multi angular approach to the research questions.

2.6 Media Literacy Index

The Media Literacy Index was introduced as an additional country-level variable during the later development of the dissertation. The index is produced by the Open Society Institute Sofia and examines the resilience of European Countries to disinformation. The 2023 index covers European countries and considers characteristics including education, media freedom, trust and opportunities for participation (Lessenski, 2023).

The inclusion of media literacy was motivated by the increasingly digital nature of political communication and participation. ECIs rely heavily on online communication and mobilisation, meaning that the ability of citizens to access, understand and critique information may potentially be relevant to political engagement (Lessenski, 2023).

The Media Literacy Index as a result, provides an additional societal indicator that could be incorporated into the existing country-level dataset. Rather than treating media literacy as a direct measure of political participation, it was considered as a potential explanatory indicator that could be investigated alongside other country-level socio-economic variables.

Exploratory analysis conducted during the dissertation revealed an interesting relationship between the Media Literacy index and signatures per capita. The direction and strength of the relationship varied significantly between individual Initiatives. Some petitions demonstrated relatively strong positive relationships, while others showed weak or negative relationships. This variation reinforced the findings from the wider literature that relationships between country-level variables and political participation may depend heavily on the specific Initiative being considered.

The Media Literacy index was consequently kept as part of the final set of variables and was incorporated into the regression, PCA and Random Forest analyses.

2.7 Research Gap

The reviewed research papers have demonstrated that significant research has been conducted on European Citizens' Initiatives, online petition participation, direct democracy and open government data. Existing studies have highlighted the importance of mobilisation strategies, campaign organisation and contextual factors in influencing participation outcomes. Previous research has also demonstrated that large scale behavioural and publicly available datasets can be analysed using quantitative methods to investigate patterns of citizen engagement (Hale et al., 2013; Lnenicka et al., 2023). However, many gaps remain.

First, reviewed literature concerning European Citizens' Initiatives focuses on democratic legitimacy, institutional effectiveness and campaign mobilisation rather than quantitative analysis of participation behaviour across multiple initiatives (Weisskircher, 2019).

Secondly, while online petition studies have demonstrated that petition data can be analysed using statistical methods, these studies are often based on national petition platforms and therefore do not examine participation across multiple European countries and Initiatives (Hale et al., 2013).

Thirdly, existing research indicates that country-level socio-economic variables can have limited explanatory power when attempting to explain political participation (Giebler et al., 2025; Geißel et al., 2023). However, limited research has examined whether combining ECI signature data with a broader range of external socio-economic indicators can provide additional explanatory power.

Finally, the literature suggests that participation can vary substantially between individual petitions because of campaign specific, issue specific or contextual reasons (Weisskircher, 2019; Giebler et al., 2025; Geißel et al., 2023). This creates an important challenge when attempting to identify relationships between country variables and participation.

This dissertation addresses these gaps by collecting ECI signature data through web scraping, integrating the resulting data with external socio-economic variables and applying a range of exploratory, statistical and modelling approaches.

The final research therefore extends beyond the initial analysis presented in the interim report. It investigates raw signature counts, participation rates across the complete dataset and participation rates when the analysis is restricted to successful Initiatives. Linear and logarithmic regression, Principal Component Analysis, Random Forest and cluster analysis are used to investigate these relationships from multiple perspectives.

The literature therefore establishes the basis for the three research questions examined in this dissertation: whether socio-economic indicators can explain raw signature counts, whether they can explain participation rates across the complete dataset, and whether they can explain participation rates when the analysis is restricted to successful ECIs.

3 Data and Methodology

3.1 Research Methodology

This dissertation follows an iterative data analytics methodology based on the principals of Cross-Industry Standard Process for Data Mining (CRISP-DM). The methodology was selected because the project involved the collection, integration, exploration and modelling of data from multiple sources, with findings from earlier stages informing subsequent analytical decisions (Data Science PM, 2024).

The overall process consisted of data collection, data preparation, exploratory data analysis, feature engineering, modelling and evaluation. Rather than just treating these stages as a linear process, the methodology was applied iteratively. Results obtained during the exploratory data analysis and initial modelling were used to refine the dataset and to inform later modelling approaches.

The initial analysis was conducted as part of the interim stage of the project. This provided an initial understanding of the dataset and identified issues that required further investigation. In particular, the exploratory analysis suggested that relationships between country-level socio-economic variables and participation were generally weak and that participation varied greatly between individual Initiatives. These findings informed the development of the final analytical approach.

The final analysis therefore expanded upon the interim methodology by introducing a second dataset containing only successful Initiatives, logarithmic transformations of the dependent variable, Principal Component Analysis (PCA), Random Forest regression and cluster analysis.

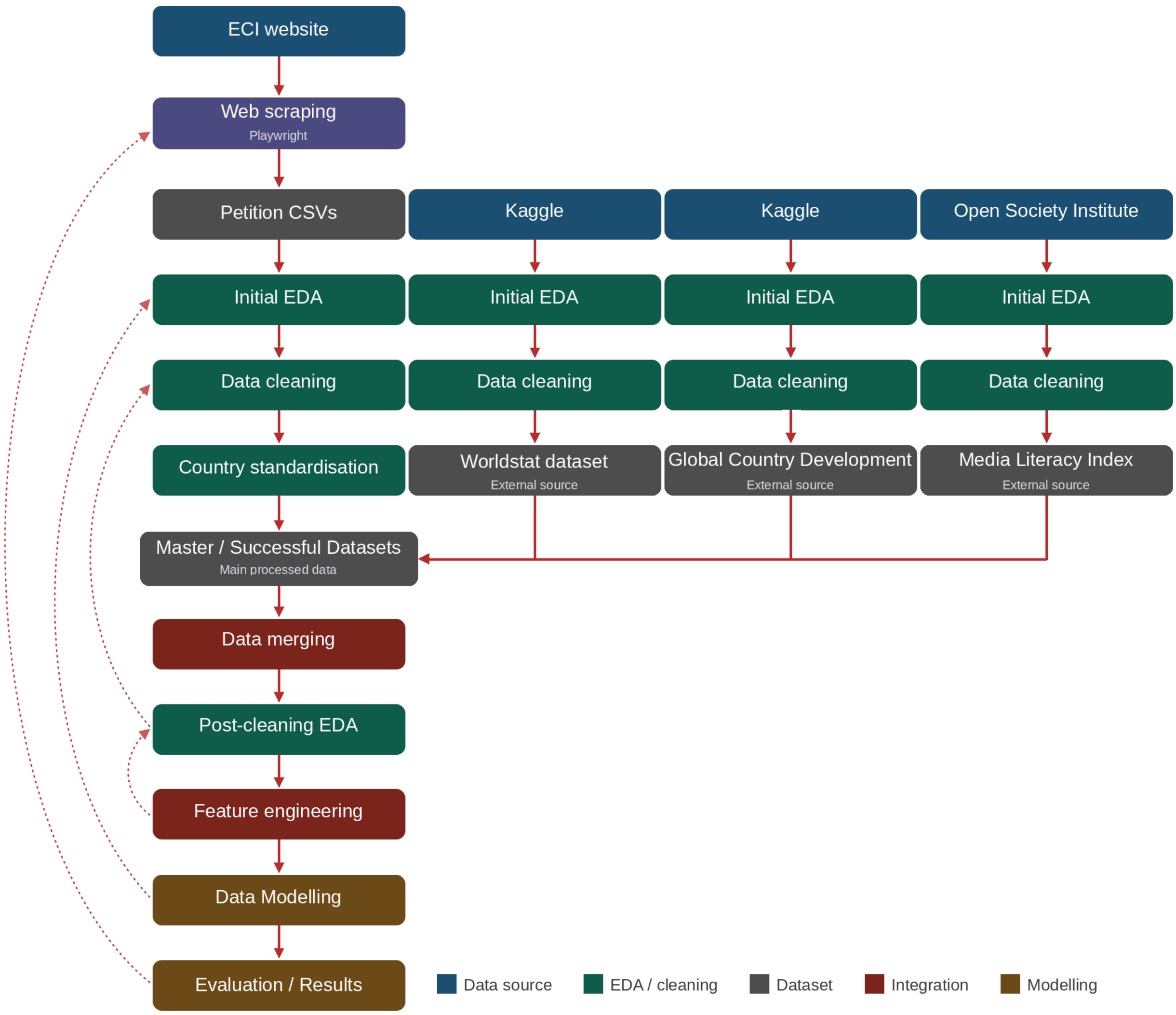

Figure 2: Project Workflow Chart

3.2 Data Collection

3.2.1 European Citizens' Initiative Data

The main data was collected from the European Citizen's Initiative platform using a custom Python web-scraping script. The scraper was developed using Python and Playwright to collect the country-level signature data manually from each individual Initiative (Medium, 2025). The scraper manually opens a chromium session on the selected page and extracts the table data from the initiative web page, extracting the country level signature data and exporting it as a CSV file.

The initital dataset contained seven ECIs:

- *Ban on Conversion Practices in the European Union* (2024)
- *European Citizens' Initiative in Defence of Agriculture and Rural Economy in Europe* (2024)
- *My Voice, My Choice: For Safe and Accessible Abortion* (2024)
- *Air-Quotas* (2024)
- *Stop Destroying Videogames* (2024)
- *Stop Cruelty Stop Slaughter* (2024)
- *Stop Fake Food: Origin on Label* (2024)

Additional Initiatives were later incorporated into the analysis, resulting in a final dataset containing 27 Initiatives. The expansion of the dataset formed part of the iterative development of the project following the interim report.

The ECI data contained information including:

- Country Name
- Number of Signatures
- Country Minimum Threshold
- Percentage of Threshold Reached

Each Initiative was exported into a separate CSV file.

3.2.2 External Socio-Economic Data

The ECI data was combined with external country-level datasets containing demographic, economic and societal indicators.

The Kaggle World Data 2023 (Elgiriyewithana, 2023) dataset was initially used as the primary source of country-level socio-economic data. The following variables from this dataset were selected:

- Country Name
- Population
- Population Density
- GDP
- Unemployment
- Tertiary Enrollment
- Life Expectancy

Additional variables were incorporated in the later stages of the project from the Media Literacy Index (Lessenski, 2023) and the 2023 Global Country Development & Prosperity Kaggle (Taşaltı̆, 2023) Dataset respectively:

- Media Literacy Index
- Governance
- Social Capital
- Personal Freedom
- Education

The inclusion of these variables allowed the analysis to investigate whether economic, demographic and societal characteristics could explain differences in participation.

3.3 Ethical Considerations

This project uses publicly available data from official European Union websites and publicly available datasets. No personally identifiable information was collected, as the ECI data used in this study is aggregated at a country level.

The web scraping process was conducted responsibly, collecting only the information required for the analysis while minimising requests to avoid placing unnecessary load on the source websites. Data was stored securely and used solely for this dissertation.

DkIT ethical approval documentation was completed prior to conducting this research. The project also followed the principles of the ACM Code of Ethics (Association for Computing Machinery, 2018), particularly regarding responsible and transparent research.

3.4 Initial Exploratory Data Analysis

Initial EDA was conducted on each dataset to examine its structure, variables, missing values, duplicate records and data types. This identified various issues requiring further cleaning, including numerical values being stored as strings, inconsistent country names and missing values within the Worldstat dataset. Each variable was also checked for summary statistics and distributions. Many of the variables especially signatures saw high levels of skewness.

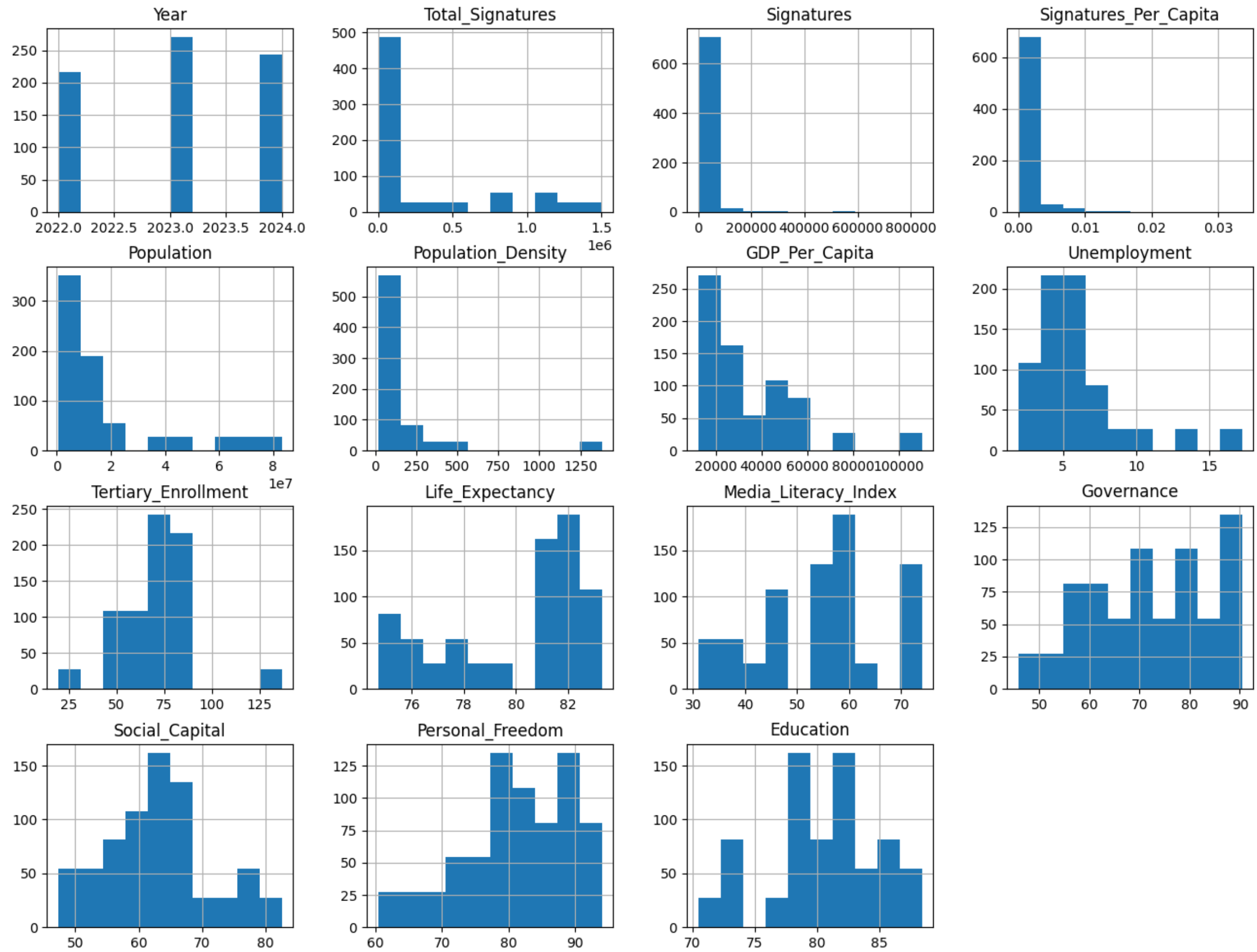

Figure 3: EDA Histograms

These findings aided in the data cleaning and preparation strategy and the distributions informed choices in feature selection and feature engineering and the modelling strategy that would follow it.

3.5 Data Preparation

The raw ECI datasets needed to undergo data preparation before they could be integrated.

First, non country rows were removed from the individual datasets. Sponsor and other information not required by the analysis were also removed. Each individual data set was checked for missing values and nonsense values, outliers and duplicate records.

Country names were then standardised so that the ECI data could be correctly matched with the external country-level datasets. This was necessary because differences in country names between datasets could result in joining issues when merged.

Numerical fields were then converted into their appropriate numerical formats. Percentage symbols, commas and currency symbols were cleaned before being converted from string into numerical values.

Following individual dataset cleaning, the Initiative datasets were combined into a single master dataset. This dataset was then merged with the external socio-economic data using using country name as the common identifier.

The same processes were employed separately with the socio-economic data before merging, such as checking for missing values, handling of string characters and conversion into numerical values.

After the interim stage, a petition specific dataset was created using the petition ID. Petition names, genre and year were manually created and a successful boolean was created and applied to each petition. This dataset was then merged with the master dataset using the petition ID.

Later in the dissertation, the Media Literacy Index and Global Country Development & Prosperity datasets were checked for missing values and the country names were standardised and merged with the master dataset before final checks and export as CSV.

As part of the analyses for the 3rd research question, a duplicate of the master dataset was created, this version of the dataset was reduced to only the successful petitions contained in the dataset using a group by of the Successful boolean.

3.6 Exploratory Data Analysis

Exploratory Data Analysis was conducted before the final modelling stage in order to identify distributions, relationships and differences between Initiatives.

Descriptive statistics and visualisations were used to examine the distributions of the main variables (Figure 11). Correlation analysis was then performed to investigate linear and monotonic relationships between participation and country-level variables (Figures 14, 15).

Both Pearson and Spearman correlations were considered. The spearman correlations were particularly useful as many variables did not appear to have straightforward linear relationships with participation, but rather non linear monotonic relationships (Figures 12, 13).

The socio-economic variables themselves shared strong correlations with one another.

The EDA also showed that population had a significantly stronger relationship with raw signature counts than many of the other socio-economic variables. However, relationships between socio-economic variables and Signatures Per Capita were generally weak.

An analysis of Media Literacy Index and signatures per capita revealed significant variance between different individual Initiatives. In some initiatives, Media Literacy had a 0.60 correlation with Signatures Per Capita, while in others it was much less significant and even negative. This further backed up the ANOVA result from the Interim report and upon further exploration between individual Initiatives, informed the creation fo the Successful Dataset as part of the third research question.

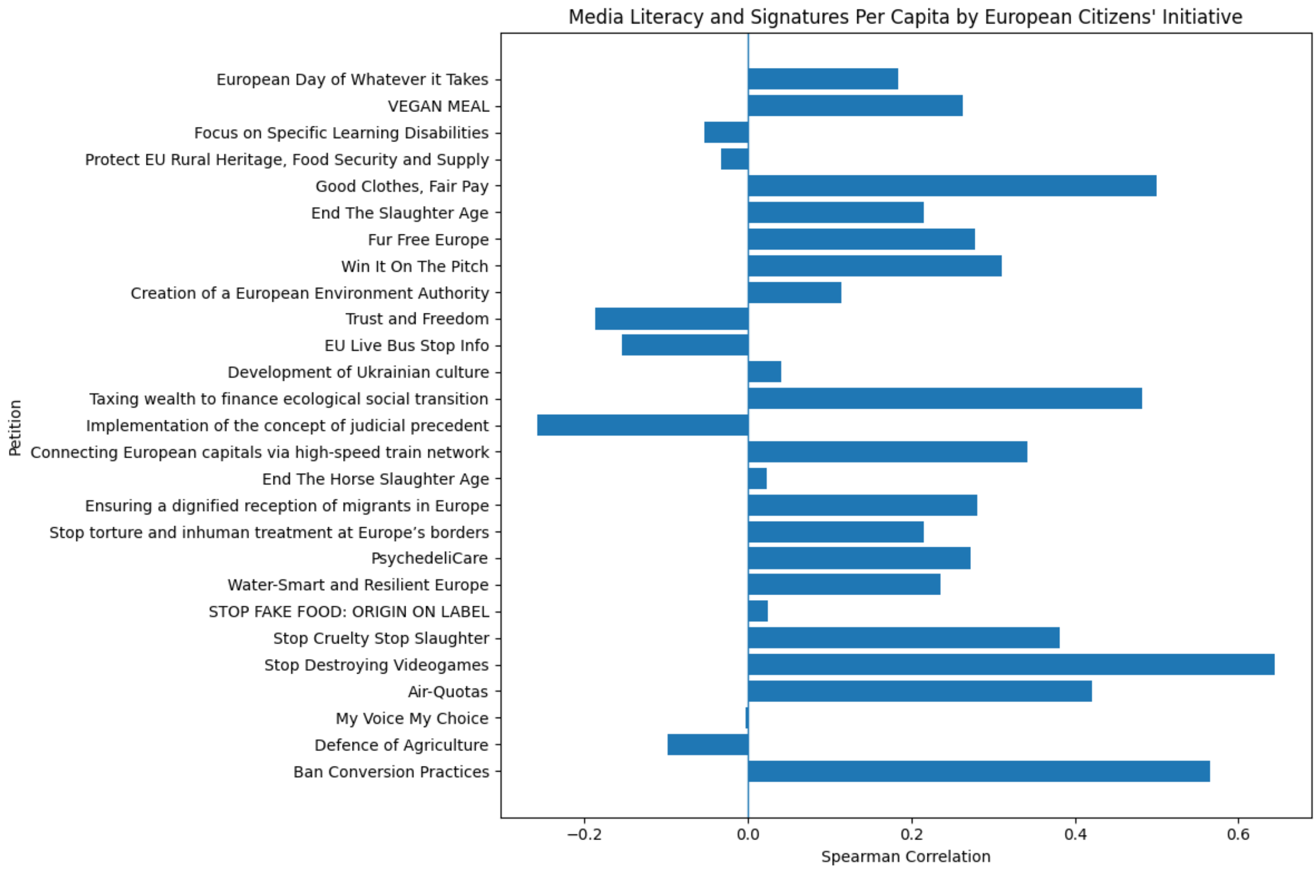

Figure 4: Correlations Signatures Per Capita vs Media Literacy grouped by Petition

These findings became important for the subsequent modelling stages as it informed some expected outcomes from the use of certain variables. For example, the multicollinear variables while weak in Linear Regressions, work very well for PCA and random forest due to the non linear nature of these models.

3.7 Feature Engineering

Features were created from the selected variables during and after the EDA stage in order to assist in creating explanatory variables as well as the main target variable for research question 2 and 3:

3.7.1 Signatures Per Capita

Signatures Per Capita was created to address differences in population size between countries. Ti was calculated by dividing the number of signatures collected in each country by the country's population. This provided a normalised measure of participation that could be used to compare participation levels between countries of different population sizes.

3.7.2 GDP Per Capita

GDP per capita was calculated by dividing the GDP of each country by its population. GDP per capita was used as a country-level socio-economic indicator within the regression and machine learning models.

3.7.3 Successful

A boolean *Successful* feature was created to identify whether an ECI was successful or unsuccessful. This feature was used to distinguish between successful and unsuccessful Initiatives and enabled the creation of the separate successful Initiatives dataset used in the later stages of the analysis.

3.7.4 Log Signatures Per Capita

A logarithmic transformation of Signatures Per Capita was created to address the highly skewed distribution observed during exploratory data analysis. The transformation reduced the influence of extremely high participation values and allowed logarithmic regression and Random Forest models to be evaluated alongside the models using the original signatures per capita values.

3.8 Feature Selection and Multicollinearity

Feature selection was performed separately for each major modelling approach rather than creating a single set of explanatory variables for the entire dissertation.

This approach was intentional as different analytical approaches have different requirements. For example, PCA benefits from multicollinear variables which can be combined into components reducing the dimensions and obtaining the underlying patterns. Linear regressions on the other hand require careful attention to multicolliearity between predictors.

Variance Inflation Factor (VIF) analysis was used during the regression modelling stages to identify multicollinearity between variables.

The feature selection process was also informed by the results of the EDA. Variables with limited relevance to the research questions or redundancy were excluded from modelling.

This resulted in multiple regression models rather than a single regression model. The approach allowed the research questions to be investigated using different combinations of socio-economic variables providing a higher level of detail and repeatability for the analyses.

3.9 Artefact Design

The artefact developed during this stage of the project is the Python based data collection, preparation, analysis and modelling pipeline.

This modular design allows for additional initiatives to be incorporated easily as the project evolves. The CRISP-DM methodology allows for iterative development meaning that stages later down the pipeline can influence reiterations in earlier stages in the pipeline.

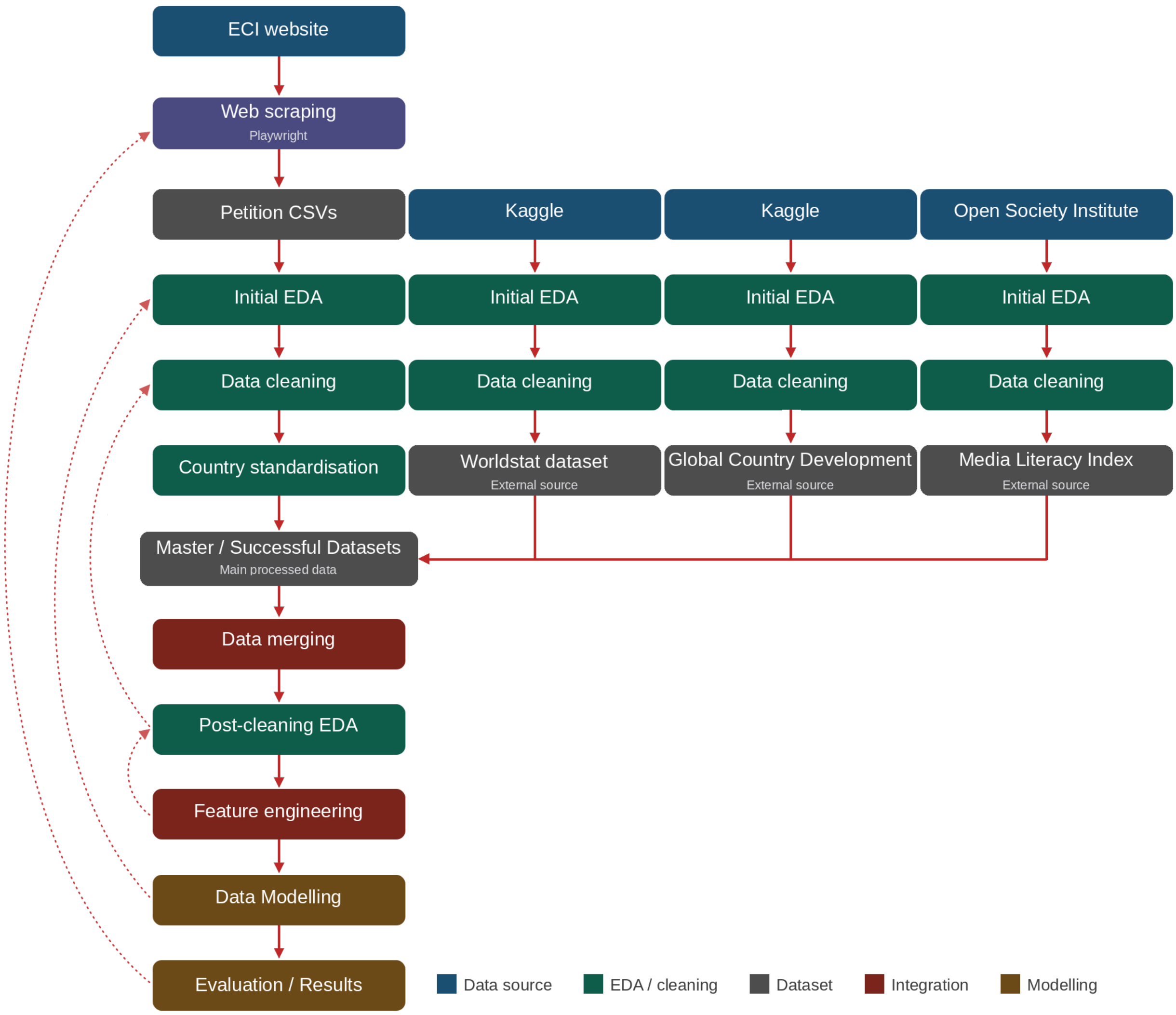

Figure 5: Project Workflow Chart

3.10 Data Modelling and Validation

3.10.1 Linear Regressions 1 (Raw Signatures)

Ordinary Least Squares (OLS) regression was used as the baseline modelling approach.

The first group of models examined whether socio-economic variables could explain raw signature counts. Three models were developed using different groups of explanatory variables.

- **Linear Regression 1A**: *Signatures Population + GDP Per Capita*
- **Linear Regression 1B**: *Signatures Population + Population Density + Tertiary Enrollment + Life Expectancy*
- **Linear Regression 1C**: *Signatures Population + Media Literacy Index + Governance + Social Capital + Personal Freedom + Education*

3.10.2 Linear Regressions 2 (Master)

The second group examined Signatures Per Capita using the complete master dataset. Three models were developed using different groups of explanatory variables.

- **Linear Regression 2A**: *Signatures Per Capita GDP Per Capita + Unemployment*
- **Linear Regression 2B**: *Signatures Per Capita GDP Per Capita + Unemployment + Tertiary Enrollment + Life Expectancy + Population Density*
- **Linear Regression 2C**: *Signatures Per Capita Media Literacy Index + Governance + Social Capital + Personal Freedom + Education*

3.10.3 Linear Regressions 3 (Successful)

A Third group repeated the Signatures Per Capita analysis using only successful Initiatives.

- **Linear Regression 3A**: *Signatures Per Capita GDP Per Capita + Unemployment*
- **Linear Regression 3B**: *Signatures Per Capita GDP Per Capita + Unemployment + Tertiary Enrollment + Life Expectancy + Population Density*
- **Linear Regression 3C**: *Signatures Per Capita Media Literacy Index + Governance + Social Capital + Personal Freedom + Education*

The models were evaluated using R-Squared, adjusted R-Squared, F-statistics, model p-values, coefficient p-values and prediction errors including Root MEAN Squared Error (RMSE) and Mean Absolute Error (MAE).

Each model was validated by checking assumptions of linearity, no multicollinearity, independence of observations, Homoscedasticity and Multivariate Normality. These assumptions were checked via VIF tests, Residual Plots and other checks. In many cases the linear regression models failed to meet all assumptions due to skewness in the variables and high degrees of multicollinearity.

3.10.4 Log Linear Regressions 2 (Master)

Because Signatures Per Capita was highly skewed, logarithmic versions of the Question 2 and Question 3 models were developed.

- **Log Linear Regression 2A**: *Log Signatures Per Capita GDP Per Capita + Unemployment*

- **Log Linear Regression 2B**: *Log Signatures Per Capita GDP Per Capita + Unemployment + Tertiary Enrollment + Life Expectancy + Population Density*

- **Log Linear Regression 2C**: *Log Signatures Per Capita Media Literacy Index + Governance + Social Capital + Personal Freedom + Education*

3.10.5 Log Linear Regressions 3 (Successful)

The logarithmic models were developed for both the complete master dataset and the successful petitions dataset. This allowed the analysis to determine whether the poor performance of the initial linear models was partly due to the highly skewed distribution of participation.

The RMSE and MAE prediction values were transformed back to their original scales to stay consistent with the other models using the numpy expm1() method. The model was then validated and evaluated used these metrics and also the Residual Plots.

- **Log Linear Regression 3A**: *Log Signatures Per Capita GDP Per Capita + Unemployment*

- **Log Linear Regression 3B**: *Log Signatures Per Capita GDP Per Capita + Unemployment + Tertiary Enrollment + Life Expectancy + Population Density*

- **Log Linear Regression 3C**: *Log Signatures Per Capita Media Literacy Index + Governance + Social Capital + Personal Freedom + Education*

3.10.6 Principal Component Analysis

PCA was applied to investigate whether correlated socio-economic variables could be represented using a smaller number of underlying dimensions.

One PCA model was created for the Master dataset and a second PCA was created for the Successful dataset, each model selected the following variables:

The variables were evaluated for multicollinearity using the Bartlett test and the VIF test. The Bartlett test achieved a significant p value, the KMO test was also employed which game a value of 0.682 meaning the variables, although mediocre, were suited for PCA.

The variables were then scaled and normalised to maintain the same scale before being included into the PCA model. Cumulative and screen plots were checked to verify the number of principal components to keep, 4 were chosen for each model to ensure it reaches 80

- Population
- Population Density
- GDP Per Capita
- Unemployment
- Tertiary Enrollment
- Life Expectancy
- Media Literacy Index
- Governance
- Social Capital
- Personal Freedom
- Education

PCA was appropriate as many of the socio-economic variables were highly correlated with one another. Rather than using all correlated variables independently within a regression model, PCA transformed the original variables into orthogonal principal components.

Four principal components were kept for the primary PCA analysis because they collectively explained 80% of the variance in the selected variables (Figures 18, 19).

The two models were then used as predictors in PCA regression models. For each PCA, a linear regression and log linear regression were created. Each model was validated using Residual Plots and evaluated using R-Squared, MAE and RMSE values.

PCA Model 01 (Master)

- **PCA Regression 01**: *Signatures Per Capita   PC1 + PC2 + PC3 + PC4*
- **PCA Log Regression 01**: *Log Signatures Per Capita   PC1 + PC2 + PC3 + PC4*

PCA Model 02 (Successful)

- **PCA Regression 02**: *Signatures Per Capita   PC1 + PC2 + PC3 + PC4*
- **PCA Log Regression 02**: *Log Signatures Per Capita   PC1 + PC2 + PC3 + PC4*

3.10.7 Random Forest Regressions

Random Forest regression was used as a machine learning approach to complement the linear statistical models.

The Random Forest models were designed to investigate whether non-linear relationships between socio-economic variables could explain participation more effectively than linear regression.

Four models were developed:

- **Random Forest 01 (Master)**: *Signatures Per Capita   Media Literacy Index + Governance + Social Capital + Personal Freedom + Education*
- **Random Forest 02 (Successful)**: *Signatures Per Capita   Media Literacy Index + Governance + Social Capital + Personal Freedom + Education*
- **Random Forest 03 (Master)**: *Log Signatures Per Capita   Media Literacy Index + Governance + Social Capital + Personal Freedom + Education*
- **Random Forest 04 (Successful)**: *Log Signatures Per Capita   Media Literacy Index + Governance + Social Capital + Personal Freedom + Education*

The data was divided into training and testing sets. Model performance was evaluated separately on the training and testing data using R-Squared. This allowed the analysis to distinguish between a model's ability to fit the training data and its ability to predict unseen observations.

3.10.8 Cluster Analysis

Finally, cluster analysis was performed as an exploratory unsupervised learning approach.

Unlike the regression models, clustering did not attempt to predict signature participation. Instead it was used to cluster groups of countries based on similar socio-economic variables.

The selected variables were standardised and clustered using K means. The elbow method was used to select the number of clusters. 4 clusters were finally selected (Figure 20).

The resulting clustered were visualised geographically using a plotly map to investigate whether groups of countries with similar socio-economic characteristics match similar patterns with countries with higher signatures per capita.

A PCA plot was also conducted to further visualise the clusters on a scatterplot (Figure 24).

4 Results, Evaluation and Comparison

4.1 Research Question 1

Can socio-economic indicators explain raw signature counts?

4.1.1 Linear Regressions

Three linear regression models were developed to investigate raw signature counts

Model 1A used GDP Per Capita and Unemployment, Model 1B added Tertiary Enrollment, Life Expectancy and Population Density and Model 1C examined Media Literacy Index, Governance, Social Capital, Personal Freedom and Education.

The models produced R-Squared values of 0.0999, 0.0994 and 0.1013 respectively. Therefore the explanatory power remained approximately 10% regardless of which group of socio-economic variables was used.

Model 1A was statistically significant overall with an F-Statistic of 40.29 and a model p-value below 0.001. However, the results were largely driven by population rather than by other socio-economic indicators, this was observed throughout the analysis including during the interim stage. The results demonstrate an important limitation of using raw signature counts. Larger countries naturally have much larger populations of potential participants. As a result, higher signature totals do not necessarily represent higher individual participation.

Table 1: Linear Regression Model Results (Question 1)

Metric	Model 1A	Model 1B	Model 1C
R Squared	0.0999	0.0994	0.1013
Adjusted R Squared	0.0974	0.0944	0.0939
F-statistic	40.2927	19.9718	13.5668
Model p-value	< 0.05	< 0.05	< 0.05
RMSE	48181.0594	48195.3254	48143.2727
MAE	16630.7002	16598.1410	16649.5806

The significance of the models mean that the Null Hypothesis for Research Question 1 gets rejected. However it is really important to note that while socio-economic variables explain some variation in raw signature counts, their explanatory power is significantly influenced by population size.

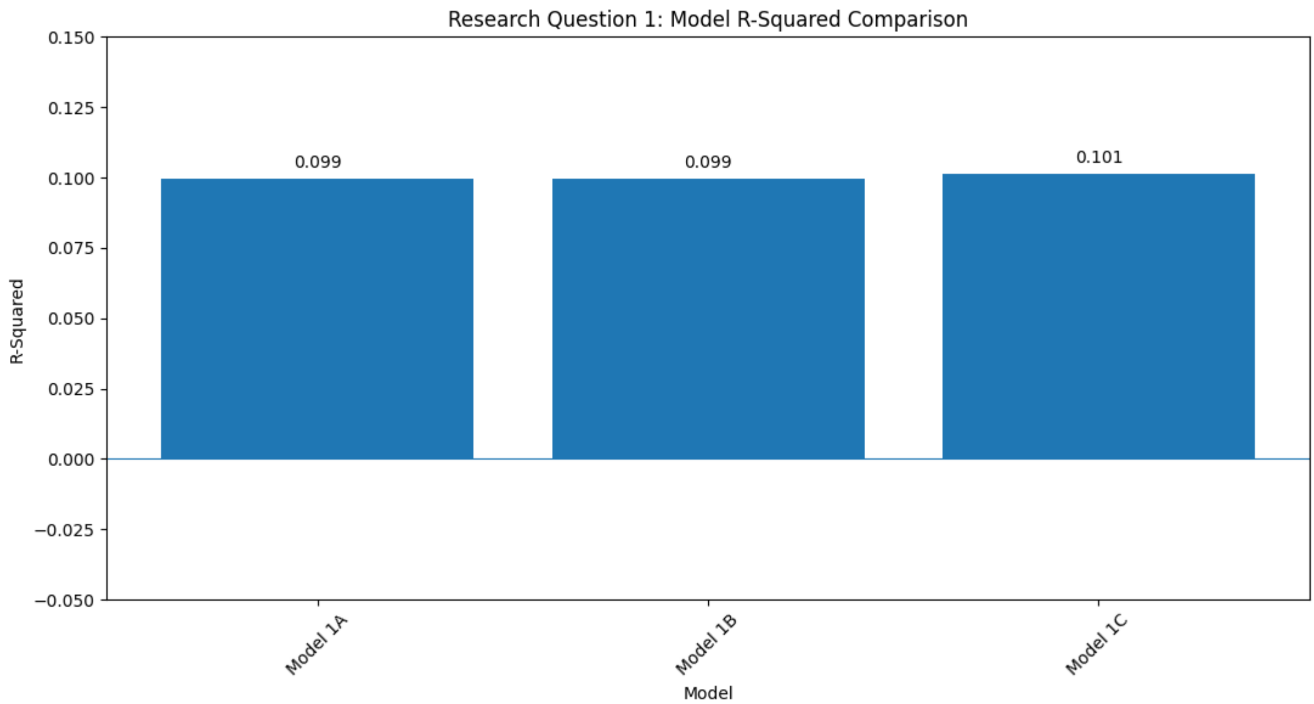

Figure 6: Question 1 R-Squared Results

4.1.2 Evaluation

The relatively similar RMSE and MAE values across the three models also indicate that adding additional socio-economic variables produced little improvement in prediction error. Model 1A produced an RMSE of 48,181.06 and MAE of 16,630.70, while Model 1c produced a slightly lower RMSE of 48,143.27 but a slightly higher MAE of 16,649.58. Overall the differences in error between the models were minimal.

These findings motivated the use of signatures per capita and the creation of Research Question 2 in order to address participation that has been normalised by population size.

4.2 Research Question 2

Can socio-economic indicators explain participation rates?

Research Question 2 examined whether socio-economic indicators could explain participation rates after accounting for population.

4.2.1 Linear Regressions

Three linear regression models were initially tested. Models 2A, 2B and 2C produced R-Squared values of 0.0045, 0.0068 and 0.0079. None of the models were statistically significant.

Table 2: Linear Regression Model Results (Question 2)

Metric	Model 2A	Model 2B	Model 2C
R Squared	0.0045	0.0068	0.0079
Adjusted R Squared	0.0018	-0.0001	0.0011
F-statistic	1.6559	0.9846	1.1566
Model p-value	0.1916	0.4262	0.3290
RMSE	0.0025	0.0025	0.0025
MAE	0.0012	0.0012	0.0012

4.2.2 Log Linear Regressions

The highly skewed distribution of Signatures Per Capita was then addressed through logarithmic transformation. However there was little change in the results. Log Models 2A, 2B and 2C produced R-Squared values of 0.0046, 0.0068 and 0.0080.

Table 3: Log Regression Model Results (Question 2)

Metric	Model 2A	Model 2B	Model 2C
R Squared	0.0046	0.0068	0.0080
Adjusted R Squared	0.0019	-0.0000	0.0012
F-statistic	1.6910	0.9962	1.1682
Model p-value	0.1851	0.4190	0.3232
RMSE	0.0025	0.0025	0.0025
MAE	0.0012	0.0012	0.0012

4.2.3 PCA Regressions

PCA was then used to examine whether relationships between correlated explanatory variables had underlying relationships that were unseen by the linear regressions. Four components were kept, representing 80% of the variance in the selected variables. The PCA regression produced an R-Squared value of 0.0089, while the Logarithmic PCA regression produced an almost identical R-Squared value of 0.0089.

Table 4: PCA Regression Model Results (Question 2)

Metric	PCA Regression 01	PCA Log Regression 01
R Squared	0.0089	0.0089
Adjusted R Squared	0.0035	0.0035
F-statistic	1.6343	1.6343
Model p-value	0.1637	0.1637
RMSE	0.0025	0.0025
MAE	0.0012	0.0012

These results indicated that the poor performance of the linear models was not simply caused by multicollinearity or the scale of Signatures per Capita.

4.2.4 Random Forest Regressions

Random Forest regression was then used to investigate whether non-linear relationships could explain variation. The normal Random Forest model achieved a training R-Squared value of 0.0492 but a testing R-Squared value of -0.0791. The Log random Forest model produced a training R-Squared value of 0.0493 and testing R-Squared value of -0.0783.

Table 5: Random Forest Model Results (Question 2)

Metric	RF Regression 01	RF Regression 03 (Log)
Training R Squared	0.0492	0.0493
Testing R Squared	-0.0791	-0.0783
Training RMSE	0.0024	0.0024
Testing RMSE	0.0028	0.0028
Training MAE	0.0011	0.0011
Testing MAE	0.0014	0.0014

4.2.5 Evaluation

The negative testing R-Squared values indicated that neither model generalised well to unseen observations. The testing RMSE and MAE were also higher than the corresponding training values, although the differences were relatively small. This indicates that the Random Forest models did not provide significant improvement over the linear approaches and were unable to produce reliable predictions for unseen observations.

4.2.6 Comparison

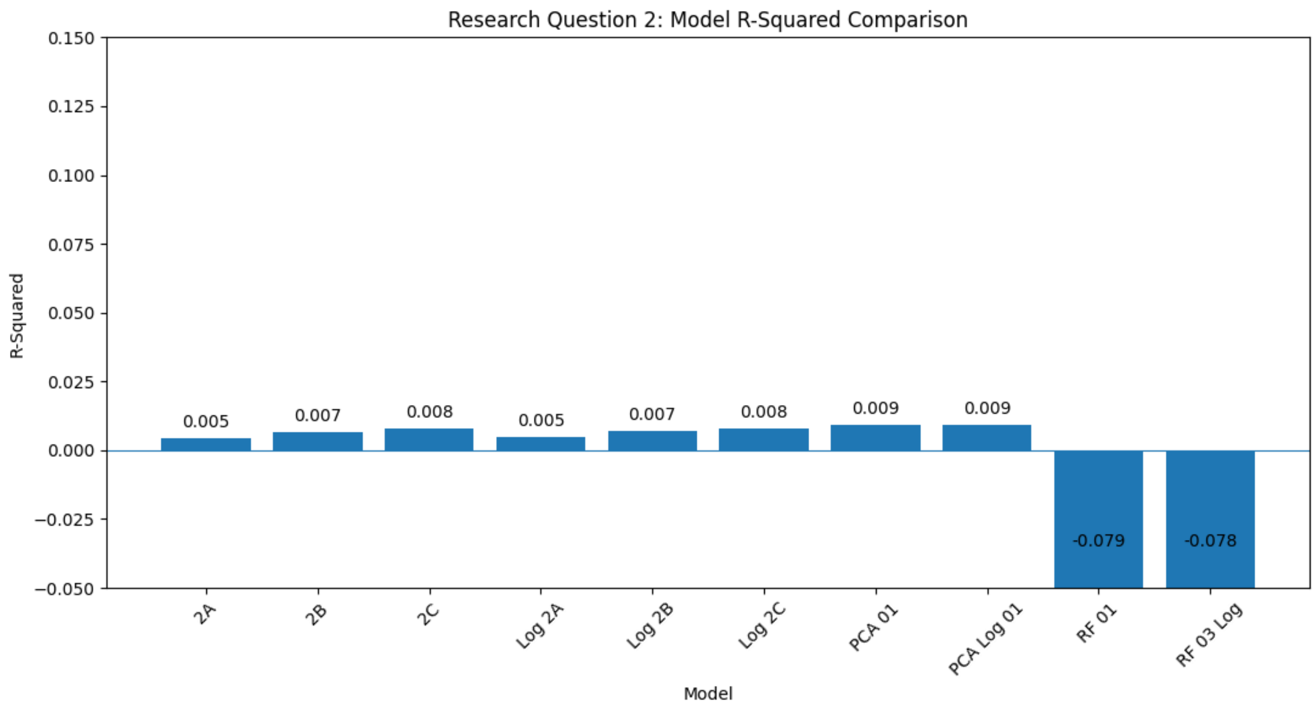

Figure 7: Question 2 R-Squared Results

Across the different modelling approaches, the results for Research Question 2 remained consistently weak. The linear, logarithmic and PCA models all produced R-Squared values below 0.01, while the Random Forest produced negative testing R-Squared values. This consistency suggests that the limited explanatory power was not specific to one modelling technique.

4.2.7 Conclusions

Overall the results provided little support for Research Question 2. The lack of significance in the models fail to reject the Null hypothesis. Socio-economic indicators demonstrated limited ability to explain participation rates across the complete dataset. This is consistent with the literature reviewed for the dissertation.

The EDA conducted revealed significant variance between petitions, especially unsuccessful petitions. The unsuccessful petitions in particular showed high degrees of unpredictability when compared to their successful counterparts. These findings informed the creation of Research Question 3 to analyse the same problem but reduced to only the successful petitions.

4.3 Research Question 3

Can socio-economic indicators explain participation rates when the analysis is reduced to only successful Petitions?

The third research question was developed to address the significant variation between Initiatives. The interim report ANOVA demonstrated significant differences in participation between Initiatives while the EDA of Media Literacy Index also showed that its relationship with Signatures Per Capita also varied significantly between individual Initiatives.

A second dataset was therefore created containing only the four successful Initiatives.

4.3.1 Linear Regressions

Table 6: Linear Regression Model Results (Question 3)

Metric	Model 3A	Model 3B	Model 3C
R Squared	0.0028	0.0329	0.0431
Adjusted R Squared	-0.0162	-0.0145	-0.0038
F-statistic	0.1490	0.6940	0.9187
Model p-value	0.8617	0.6291	0.4721
RMSE	0.0049	0.0049	0.0048
MAE	0.0029	0.0029	0.0029

The three linear regression models produced R-Squared values of 0.0028, 0.0329 and 0.0431. Although these were slightly higher than the corresponding models from research question 2, none of the models were statistically significant.

4.3.2 Log Linear Regressions

Table 7: Log Regression Model Results (Question 3)

Metric	Model 3A	Model 3B	Model 3C
R Squared	0.0560	0.0788	0.1187
Adjusted R Squared	0.0380	0.0337	0.0755
F-statistic	3.1115	1.7460	2.7466
Model p-value	0.0487	0.1308	0.0227
RMSE	0.0052	0.0051	0.0051
MAE	0.0025	0.0025	0.0024

The logarithmic models produced stronger results. Log Model 3A achieved and R-Squared of 0.0560 and was statistically significant with a model p-value of 0.0487. Log Model 3B achieved an R-Squared of 0.0788, although its p-value of 0.131 was not significant. Log Model 3C produced the strongest linear regression result, with an R-Squared of 0.1187, adjusted R-squared of 0.0755 and model p-value of 0.0227.

4.3.3 PCA Regressions

Table 8: PCA Regression Model Results (Question 3)

PCA Regression 02	PCA Log Regression 02	
R Squared	0.0679	0.1341
Adjusted R Squared	0.0317	0.1004
F-statistic	1.8771	3.9865
Model p-value	0.1201	0.0048
RMSE	0.0048	0.0050
MAE	0.0028	0.0024

The PCA analysis also showed strong improvement. The non-logarithmic PCA regression produced an R-Squared value of 0.068, while the logarithmic PCA regression produced an R-Squared of 0.1341, R-Squared adjusted of 0.1004 and a model p-value of 0.00477.

4.3.4 Random Forest Regressions

Table 9: Random Forest Model Results (Question 3)

Metric	**RF Regression 02**	**RF Regression 04 (Log)**
Training R Squared	0.3153	0.3553
Testing R Squared	-0.8275	-0.0207
Training RMSE	0.0044	0.6968
Testing RMSE	0.0039	1.1957
Training MAE	0.0026	0.5598
Testing MAE	0.0029	0.9555

The Random Forest results were more mixed. The normal successful dataset model achieved a training R-Squared of 0.3153 but a testing R-Squared of -0.8275. This indicates strong levels of overfitting with the training data. The logarithmic version improved considerably with a training R-Squared of 0.3553 and a testing R-Squared of -0.0207, although the testing performance still did not demonstrate strong predictive ability.

4.3.5 Evaluation

The other evaluation metrics also demonstrate the difference between the normal and logarithmic Random Forest models. The normal model produced a testing RMSE of 0.0039 and MAE of 0.0029, while the logarithmic model produced significantly larger testing errors of 1.1957 and 0.9555. However, these error values are based on the transformed target for the logarithmic model and should therefore not be directly compared with the non-logarithmic errors.

4.3.6 Comparison

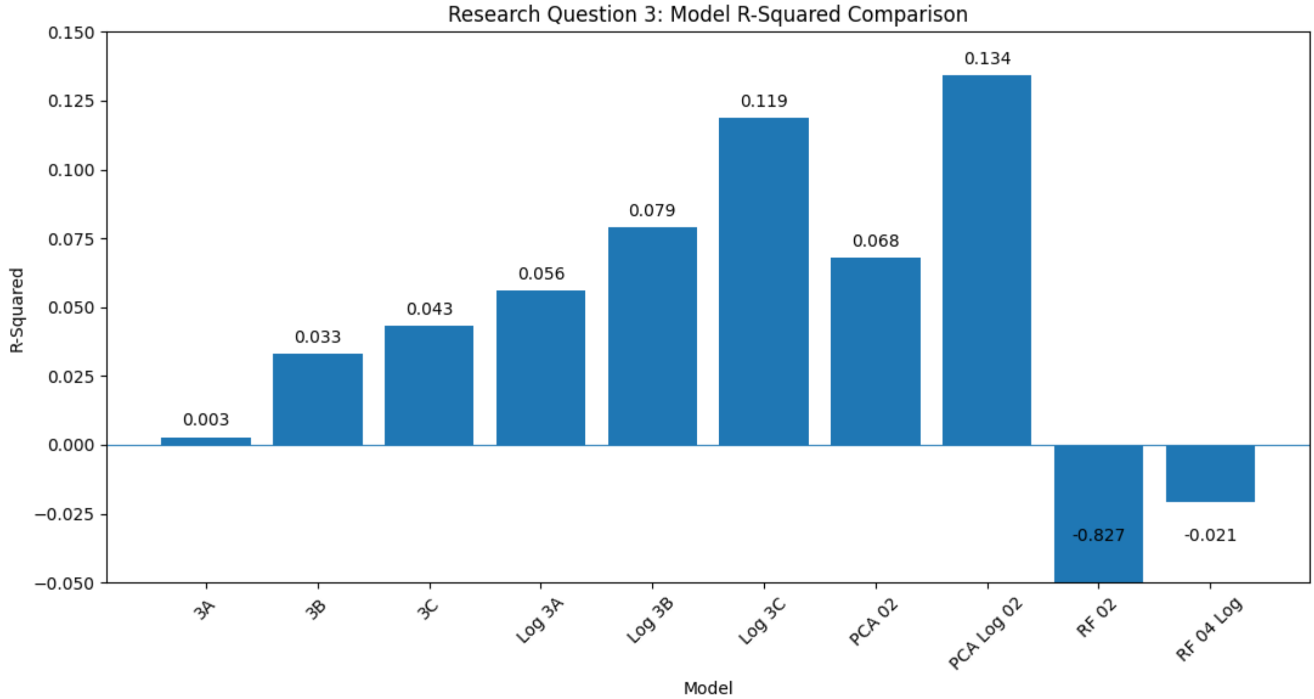

Figure 8: Question 3 R-Squared Results

Comparing the results across Research Question 2 and 3 shows that restricting the analysis to successful Inititatives produced stronger model performance. The largest improvement was observed in the logarithmic PCA regression which achieved an R-Squared of 0.1341 and an adjusted R-Squared of 0.1004. However the Random Forest results remained inconsistent, demonstrating that improved explanatory performance did not necessarily translate into strong predictive performance.

4.3.7 Conclusions

Overall, Research Question 3 obtained much stronger results than Research Question 2. Restricting the analysis to successful Initiatives produced stronger relationships, especially when logarithmic transformations and PCA were used. However, the relatively small successful dataset limits the reliability of these findings. The significant models Log

3A, Log 3B and Log PCA regression 02 reject the Null Hypothesis for research Question 3. Country-level socio-economic variables can explain a proportion of the variance in signatures per capita when the analysis is restricted to only successful petitions, and the target variable has underwent a logarithmic transformation.

4.4 Cluster Analysis

Cluster analysis was conducted as an exploratory approach rather than as a predictive model. The clusters were created using K-means clustering which groups unlabeled data based on feature similarities of the selected country-level socio-economic variables. The elbow method was used to decide the number of clusters to keep. 4 clusters were chosen for this analysis (Figure 20).

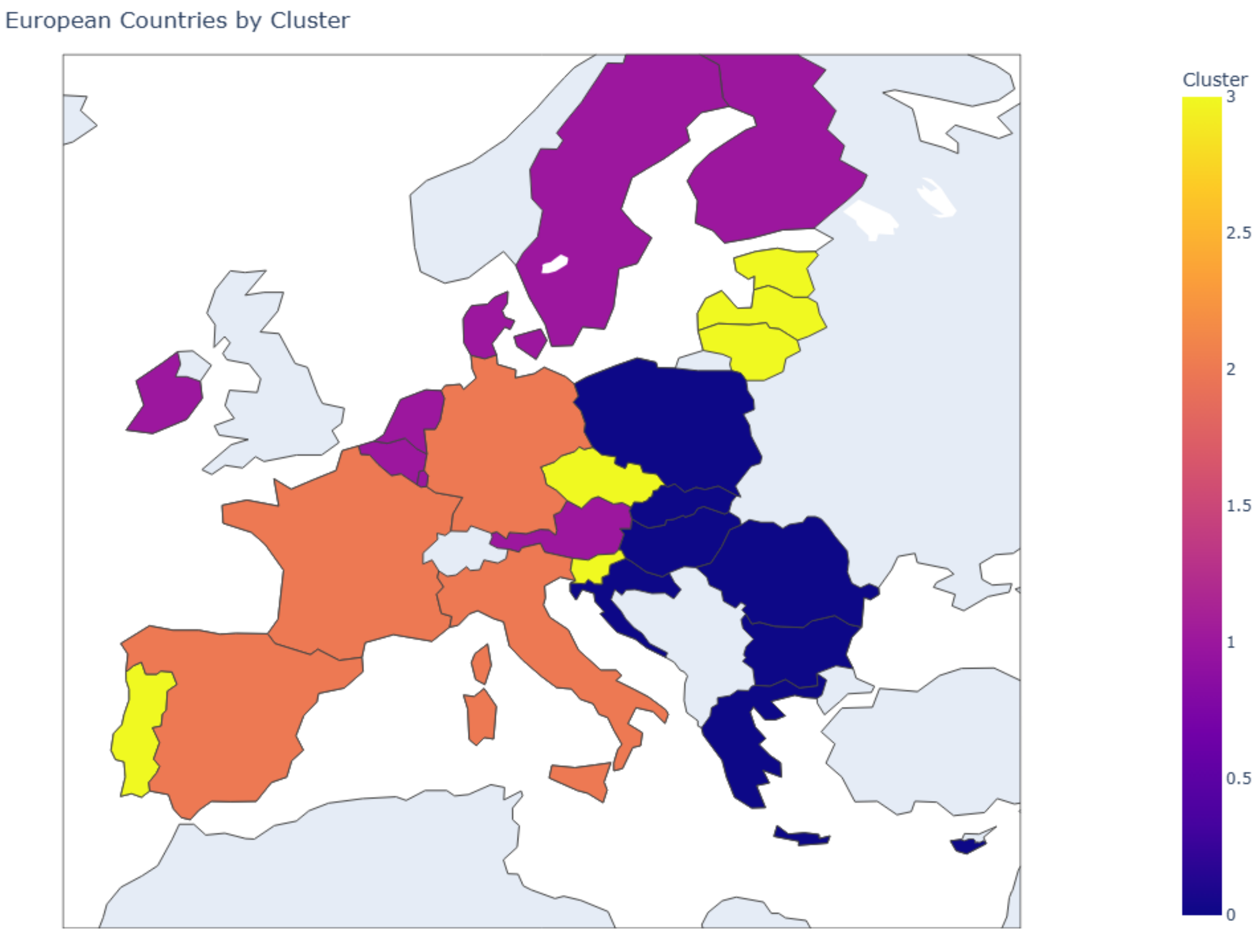

Figure 9: Country Clusters Map

The clusters were compared against Signatures Per Capita and Log Signatures Per Capita using boxplots. The boxplots revealed significant variation across clusters (Figures 21, 22). The resulting clusters were also subsequently visualised geographically using a plotly map to investigate whether countries with similar socio-economic characteristics formed identifiable regional patterns.

A PCA of the selected variables was also created to highlight the clusters in a scatter plot. It further showed distinct clusters in the analysis (Figure 24).

The cluster analysis did not demonstrate that country clusters directly determined participation. Instead, it provided additional evidence that countries can be grouped according to similarities in their socio-economic characteristics, offering a descriptive perspective that complemented the regression analysis.

It is also worth noting that the highlighted country clusters highlighted similar patterns to the country clusters in the Media Literacy Index by (Lessenski, 2023). It also showed similar patterns to country signatures across many individual petitions, for example, the Nordic countries and Ireland are in the same cluster which all show similar levels of participation upon individual inspection of the Initiatives.

4.5 Overall Findings

Across the analysis, the strongest finding is that country-level socio-economic indicators alone provide a limited proportion of explanatory power for ECI participation.

Raw signature counts showed stronger relationships with population, but these relationships largely disappeared when participation was normalised by population. Neither linear regression, logarithmic regression, PCA nor Random Forest produced strong predictive performance across the whole dataset that included all petitions.

The successful Initiative dataset analysis produced much stronger results, especially for Logarithmic PCA regression, but the smaller sample size and poor Random Forest generalisation meant that these findings should be interpreted with caution.

The results therefore suggest that ECI participation is largely influenced by factors beyond just the socio-economic characteristics examined in this dissertation. The significant variation between Initiatives identified through the ANOVA, together with the differing relationships observed between individual petitions, supports the possibility that campaign strategy, political context, moblisation and other Initiative specific characteristics play a much more substantial role in explaining variance in participation rates, as confirmed by the findings of the reviewed literature.

Consequently, the dissertation finds that socio-economic indicators may contribute a to explaining participation under certain conditions, but they cannot adequately explain ECI participation on their own. In a greater analysis, socio-economic indicators would be included to explain a significant but small fraction of the variance, combined with issue specific and campaign specific factors to get a more complete picture.

5 Conclusion and Future Work

5.1 Conclusions

This dissertation investigated whether country-level socio-economic characteristics could explain participation in European Citizens' Initiatives (ECIs). The project followed an iterative CRISP-DM approach, progressing from web scraping and data preparation through exploratory data analysis, statistical modelling, machine learning and cluster analysis.

The first research question examined whether socio-economic indicators could explain raw signature counts. The three linear regression models achieved R-Squared values of approximately 0.10, with population being the most influential variable compared to the other socio-economic variables. This demonstrated the limitation of using raw signature counts, as larger countries naturally have larger potential populations of participants.

The second research question therefore examined whether socio-economic indicators could explain variation in participation rates, using Signatures Per Capita. Linear, logarithmic and PCA regression models all produced very low explanatory power when applied to the complete master dataset. Random Forest regression also failed to generalise effectively to unseen observations. These results provide little evidence that country-level socio-economic characteristics alone can explain participation rates.

The third research question examined whether relationships changed when the analysis was restricted to successful Initiatives. Some improvement was observed, especially following logarithmic transformation and PCA. The logarithmic PCA regression achieved an R-Squared of 0.134 and was statistically significant overall. However, the Random Forest results demonstrated strong overfitting, especially for the non-logarithmic model. Therefore, the improved performance should be interpreted with caution rather than as evidence of strong predictive capability.

Overall the findings indicate that socio-economic characteristics may contribute to participation but cannot adequately explain participation across ECIs on their own. This is consistent with the reviewed literature, especially from Weisskircher (2019) and Giebler et al. (2025), who identified mobilisation strategies, campaign organisation, political context and issue specific factors as potentially more important influences.

The dissertation therefore demonstrates the importance of considering participation as a multi-factored issue rather than attempting to explain it through a limited set of country-level variables.

5.2 Future Work

The most important area for future work would be to increase the number of initiatives included in the dataset. A larger dataset covering more ECIs over a longer period would allow country-level and Initiative level effects to be examined more effectively. This would address the limitations of the successful dataset lacking significant depth in Research Question 3.

Another area for future work would be fine tuning of machine learning models and possibly employing more complex models apt for the analysis. The Random Forest Models were incapable of capturing the variance in the test due to the lack of depth of the dataset at hand, and also due to the significant overfitting of the dataset. Training the model with a higher epoch and combined with a larger dataset would help to alleviate the limitations of the models used in the dissertation.

Further analysis into areas beyond country-level socio economic variables such as issue specific factors. An attempt was made to expand into this area through feature engineering of a petition genre categorical variable to analyse the influence of petition genres on participation. The limited extent of the dataset and avoiding scope creep resulted in this variable remaining unused. A future analysis could further incorporate this into the analysis as well as find more petition specific and issue specific variables.

Finally, a significant development in the project would be the adoption of a Natural Language Processing model to analyse consensus on social media platforms, this branch off was too high an expansion of scope for the current dissertation, but future work could expand into this area so that a proper analysis could be made on campaign-level factors. All of these factors could then be combined with the transformed socio-economic variables to obtain a more complete analysis of variance in participation suing a multi-factored approach.

Bibliography

Air-quotas. (2024). https://citizens-initiative.europa.eu/initiatives/details/2024/000006_en.

Association for Computing Machinery. (2018). *Acm code of ethics and professional conduct.* Retrieved from https://www.acm.org/code-of-ethics (Accessed: 07 April 2026)

Ban on conversion practices in the European Union. (2024). https://citizens-initiative.europa.eu/initiatives/details/2024/000001_en.

Data Science PM. (2024). *What is crisp dm? data science project management.* https://www.datascience-pm.com/crisp-dm-2/.

Elgiriyewithana, N. (2023). *Global country information dataset 2023.* Data set. Kaggle. Retrieved from https://doi.org/10.34740/KAGGLE/DSV/6101670 doi: 10.34740/KAGGLE/DSV/6101670

Europa.eu. (2024). *Thresholds.* European Citizens' Initiative. Retrieved from https://citizens-initiative.europa.eu/thresholds_en

European Commission. (n.d.). *History of the european citizens' initiative.* https://citizens-initiative.europa.eu/how-it-works/history_en.

European citizens' initiative in defence of agriculture and rural economy in Europe. (2024). https://citizens-initiative.europa.eu/initiatives/details/2024/000002_en.

Geißel, B., Krämling, A., & Paulus, L. (2023). *Direct democracy and equality: context is the key.* https://doi.org/10.1057/s41269-023-00316-4.

Giebler, H., Giesecke, J., Humphreys, M., Hutter, S., & Klüver, H. (2025). *Mobilizing europe's citizens to take action on migration and climate change: behavioral evidence from 27 eu member states.* https://doi.org/10.1080/13501763.2025.2512032.

Hale, S. A., Margetts, H., & Yasseri, T. (2013). *Petition growth and success rates on the uk no. 10 downing street website.* https://doi.org/10.1145/2464464.2464518.

Jensen, K. (2012). *Crisp-dm process diagram.* Wikimedia Commons. Retrieved from https://commons.wikimedia.org/wiki/File:CRISP-DM_Process_Diagram.png (Licensed under Creative Commons Attribution-ShareAlike 3.0)

Lessenski, M. (2023). *Finland tops the new media literacy index 2023, countries close to the war in ukraine remain among the most vulnerable to disinformation.* https://osis.bg/?p=4450&lang=en.

Lnenicka, M., Nikiforova, A., Luterek, M., Milic, P., Rudmark, D., Neumaier, S., … Pedro, M. (2023). *Identifying patterns and recommendations of and for sustainable open data initiatives: a benchmarking-driven analysis of open government data initiatives among european countries.* https://arxiv.org/abs/2312.00551.

Medium. (2025). *How to scrape websites with Playwright and Python.* https://medium.com/@hasdata/how-to-scrape-websites-with-playwright-and-python-49a015fd00aa.

My voice, my choice: For safe and accessible abortion. (2024). https://citizens-initiative.europa.eu/initiatives/details/2024/000004_en.

Stop cruelty stop slaughter. (2024). https://citizens-initiative.europa.eu/initiatives/details/2024/000008_en.

Stop destroying videogames. (2024). https://citizens-initiative.europa.eu/initiatives/details/2024/000007_en.

Stop fake food: Origin on label. (2024). https://citizens-initiative.europa.eu/initiatives/details/2024/000009_en.

Taşaltĭ, T. T. (2023). *2023 global country development & prosperity index.* Data set. Kaggle. Retrieved from https://www.kaggle.com/datasets/tarktunataalt/2023-global-country-development-and-prosperity-index

Weisskircher, M. (2019). *The european citizens' initiative: Mobilization strategies and consequences.* https://doi.org/10.1177/0032321719859792.

A Appendices

A.1 Project Workflow Chart

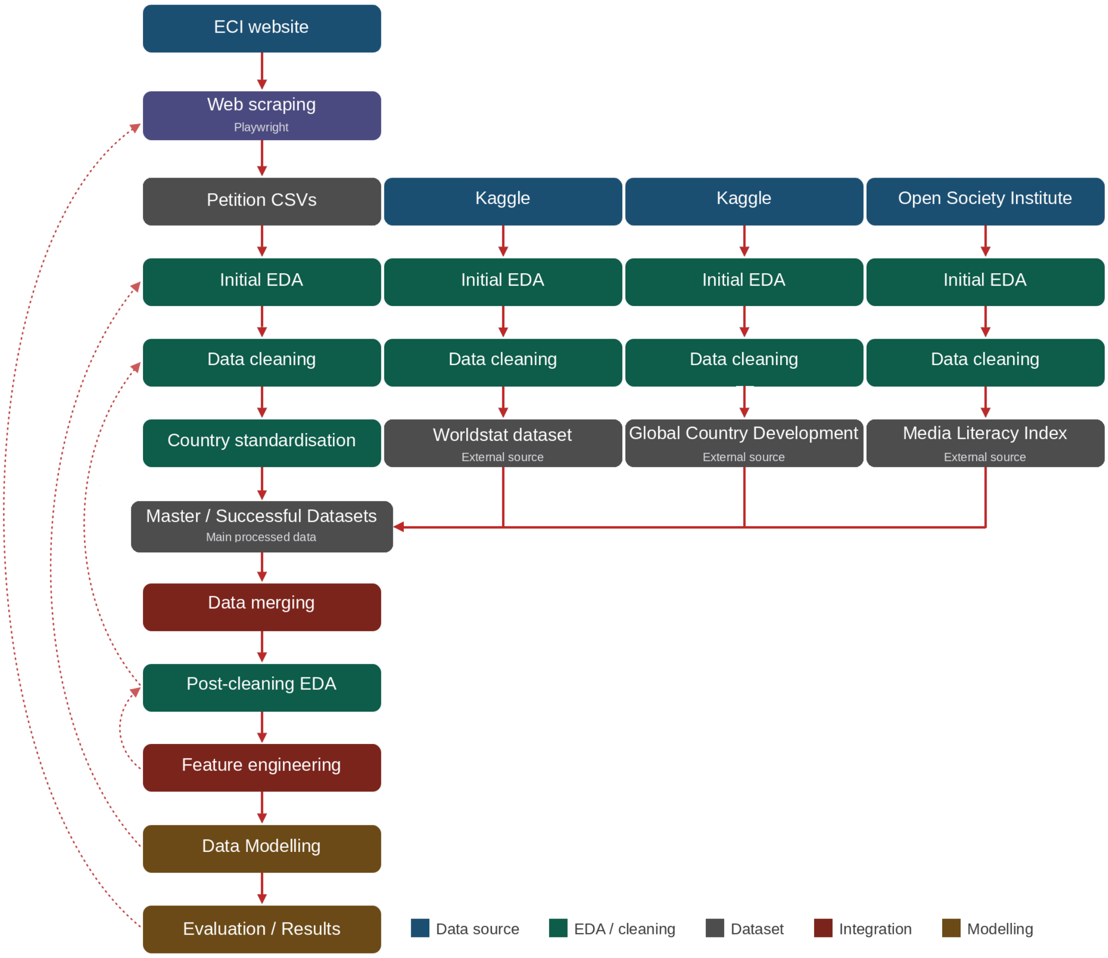

Figure 10: Project Workflow Chart

A.2 EDA Histograms

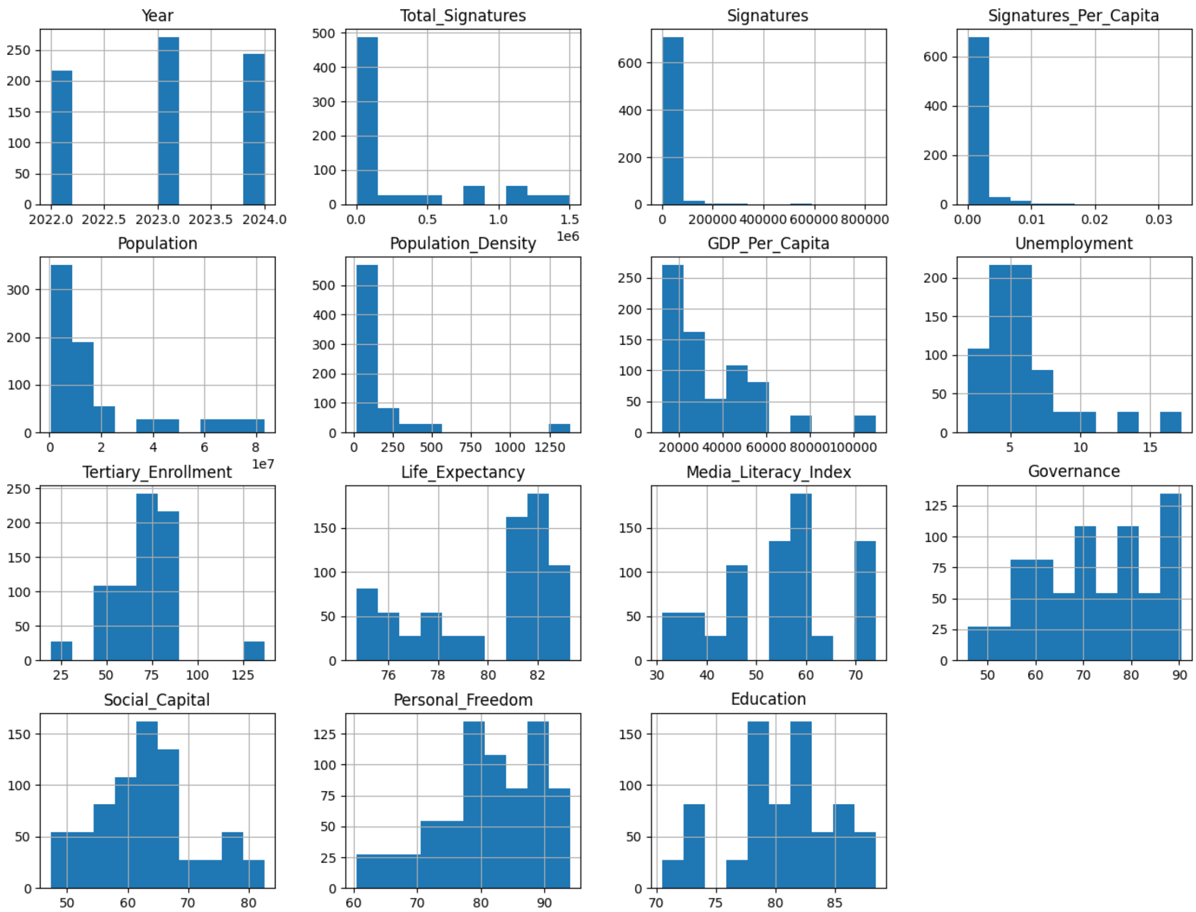

Figure 11: EDA Histograms

A.3 EDA Spearman Heatmap (Master Dataset)

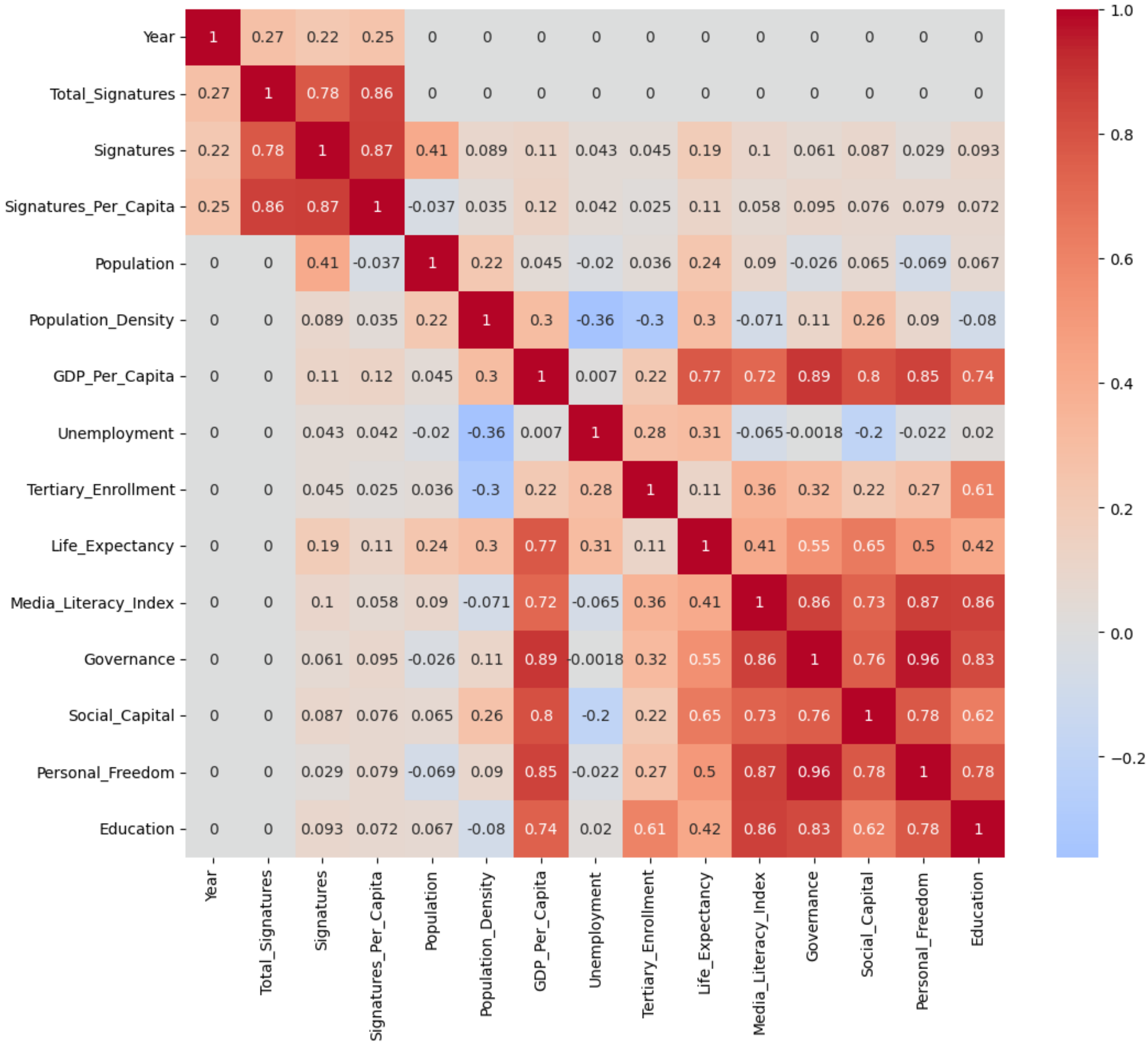

Figure 12: Spearman Heatmap for the Master Dataset

A.4 EDA Spearman Heatmap (Successful Dataset)

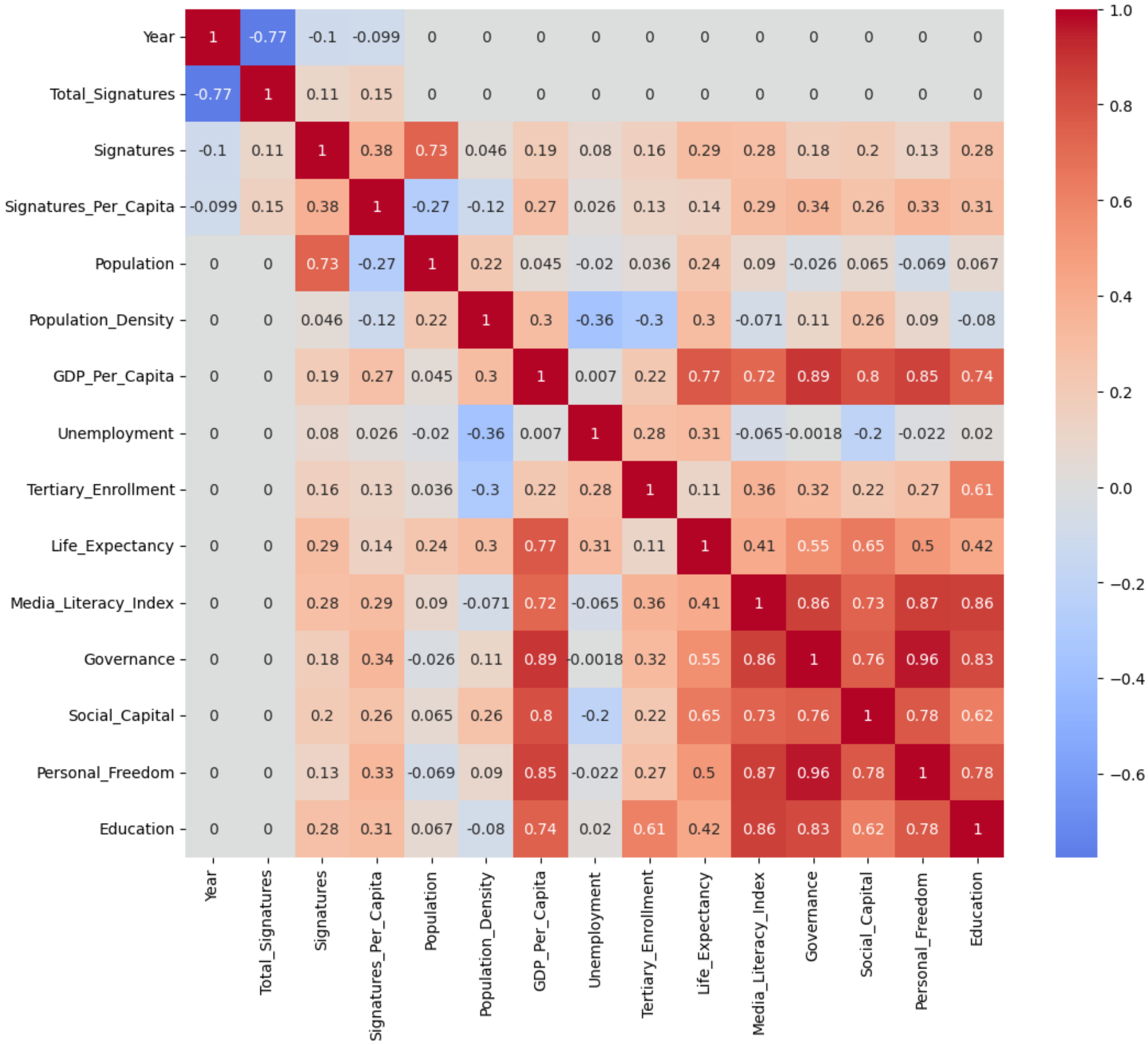

Figure 13: Spearman Heatmap for the Successful Dataset

A.5 EDA Scatterplots (Master Dataset)

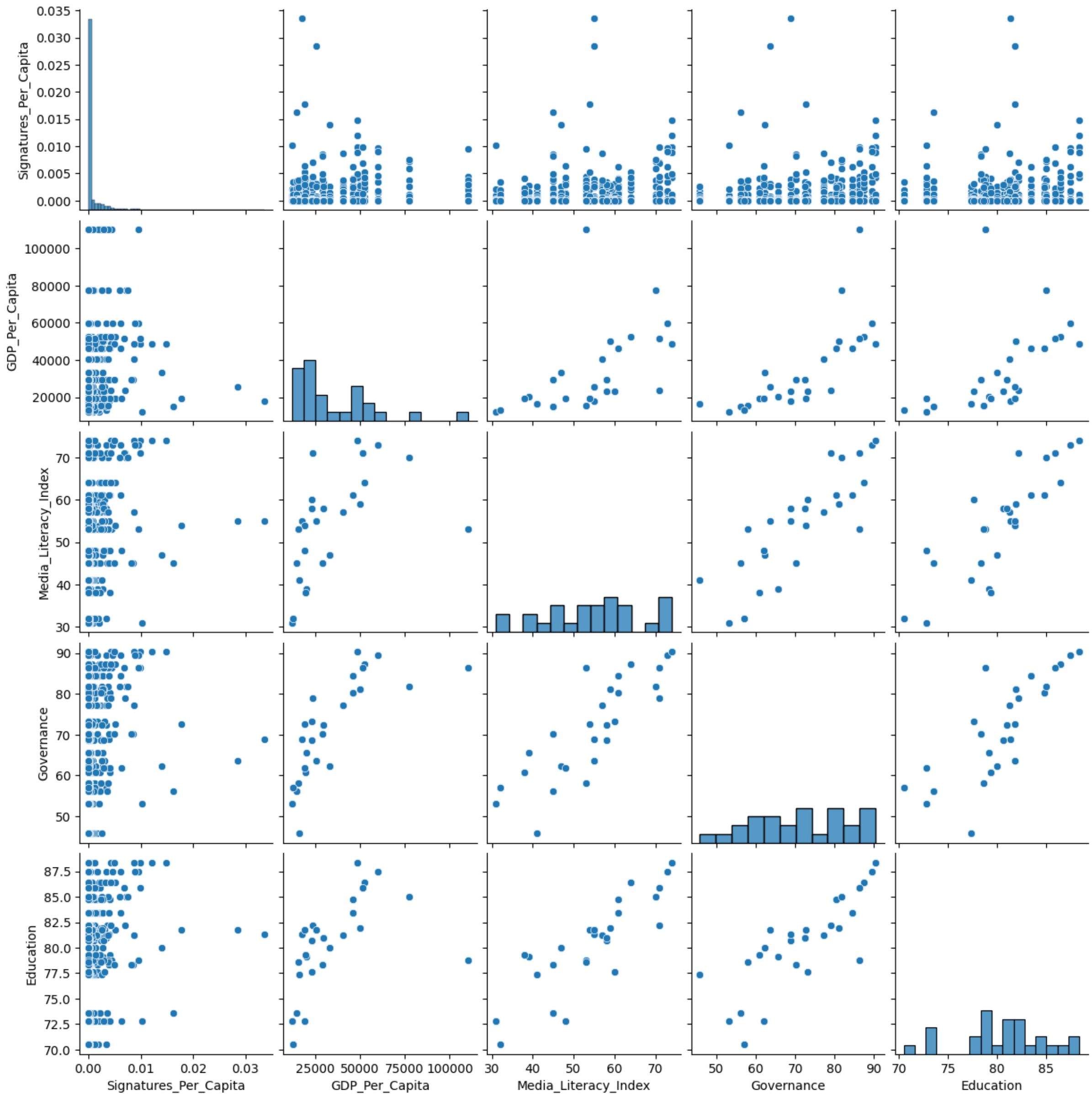

Figure 14: Pairplots for the Master Dataset

A.6 EDA Scatterplots (Successful Dataset)

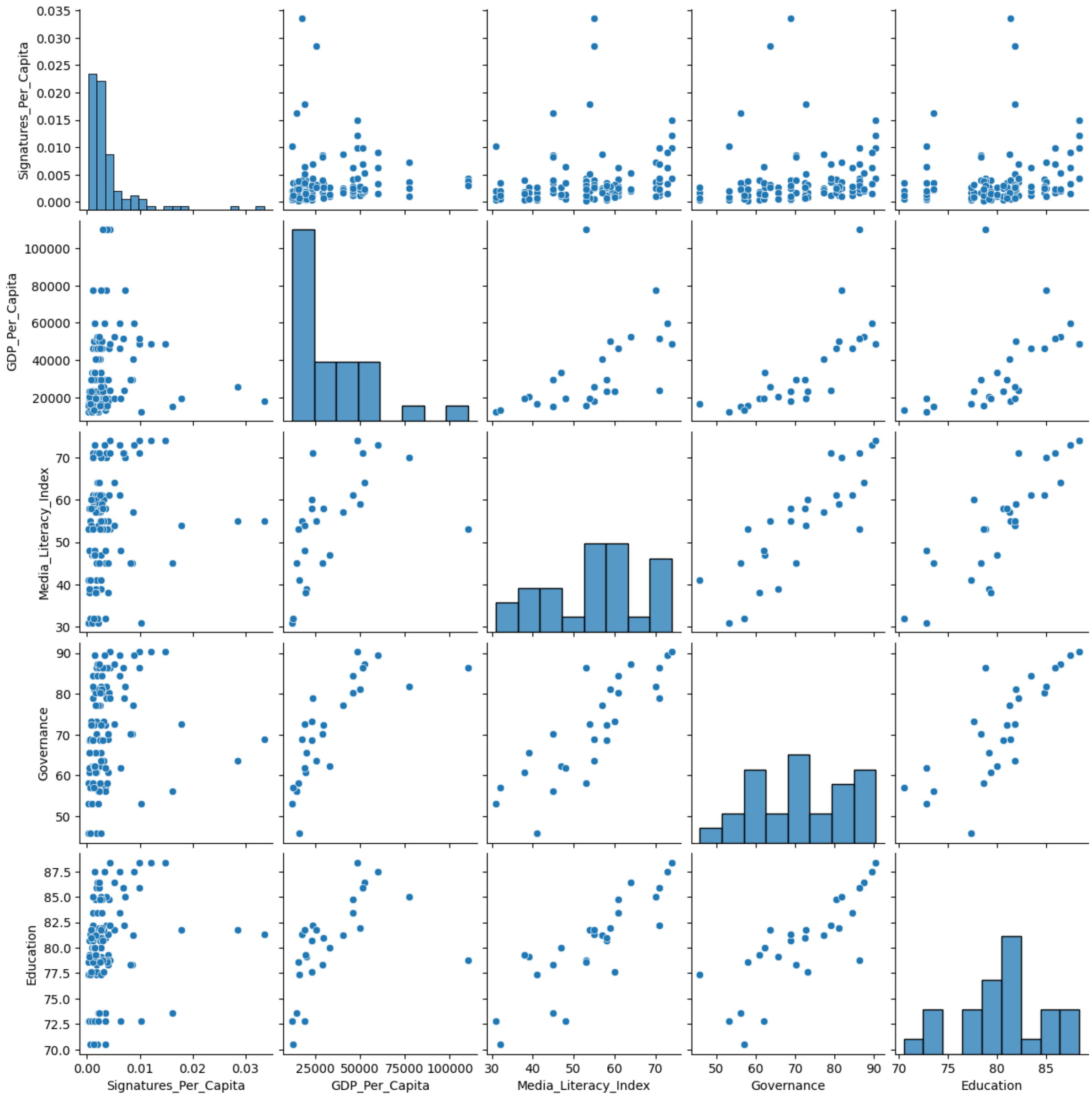

Figure 15: Pairplots for the Successful Dataset

A.7 Countries with highest Signatures per Capita

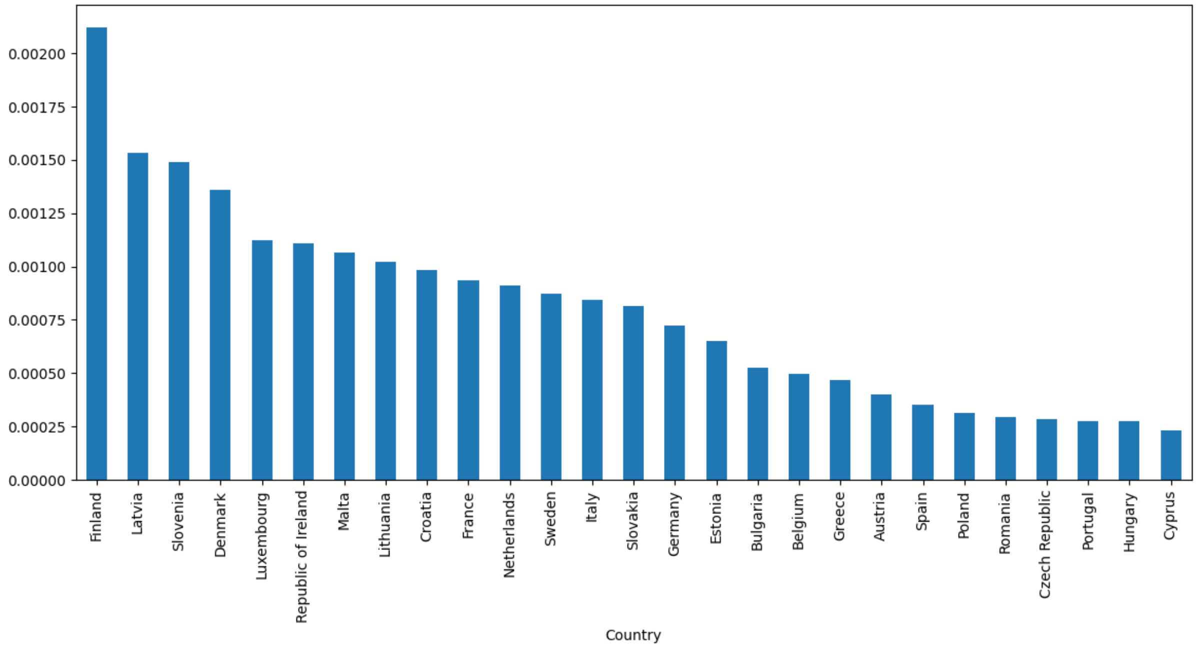

Figure 16: Countries with highest Signatures per Capita

A.8 Correlations between Signatures Per Capita and Media Literacy Index

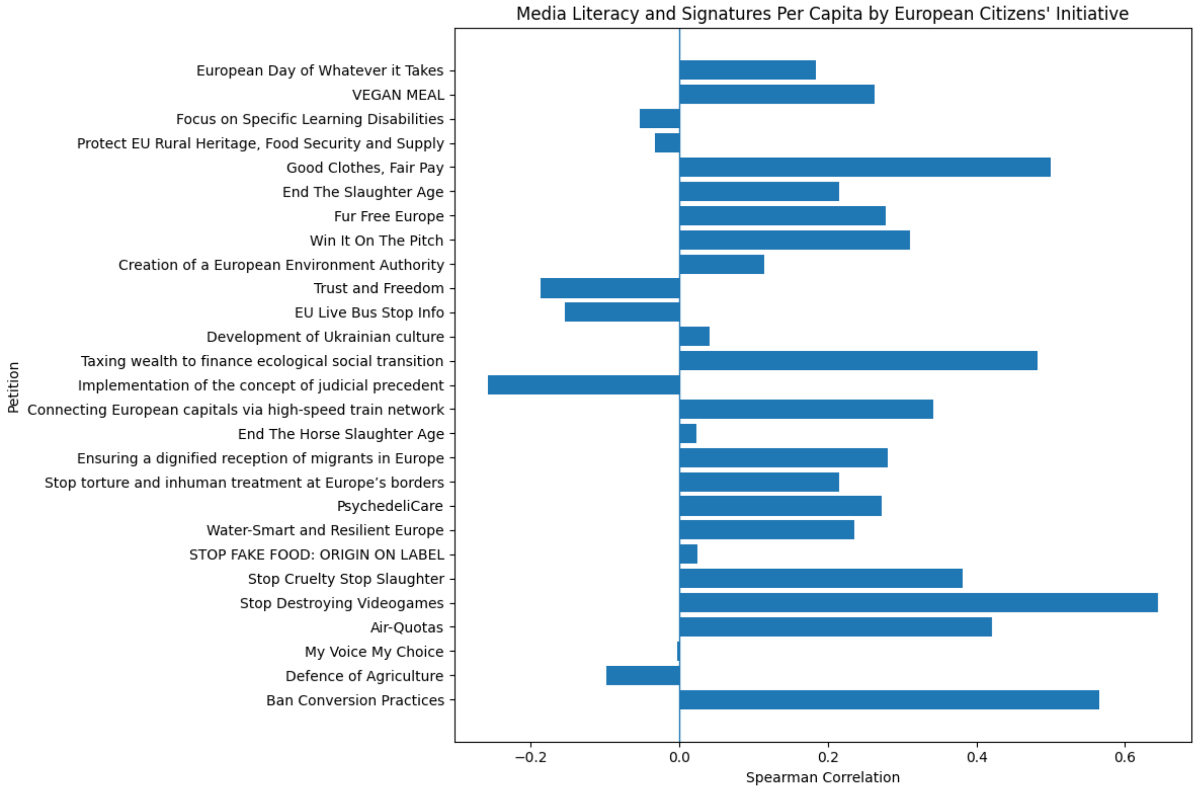

Figure 17: Correlations Signatures Per Capita vs Media Literacy grouped by Petition

A.9 PCA 01 Scree Plot (Master Dataset)

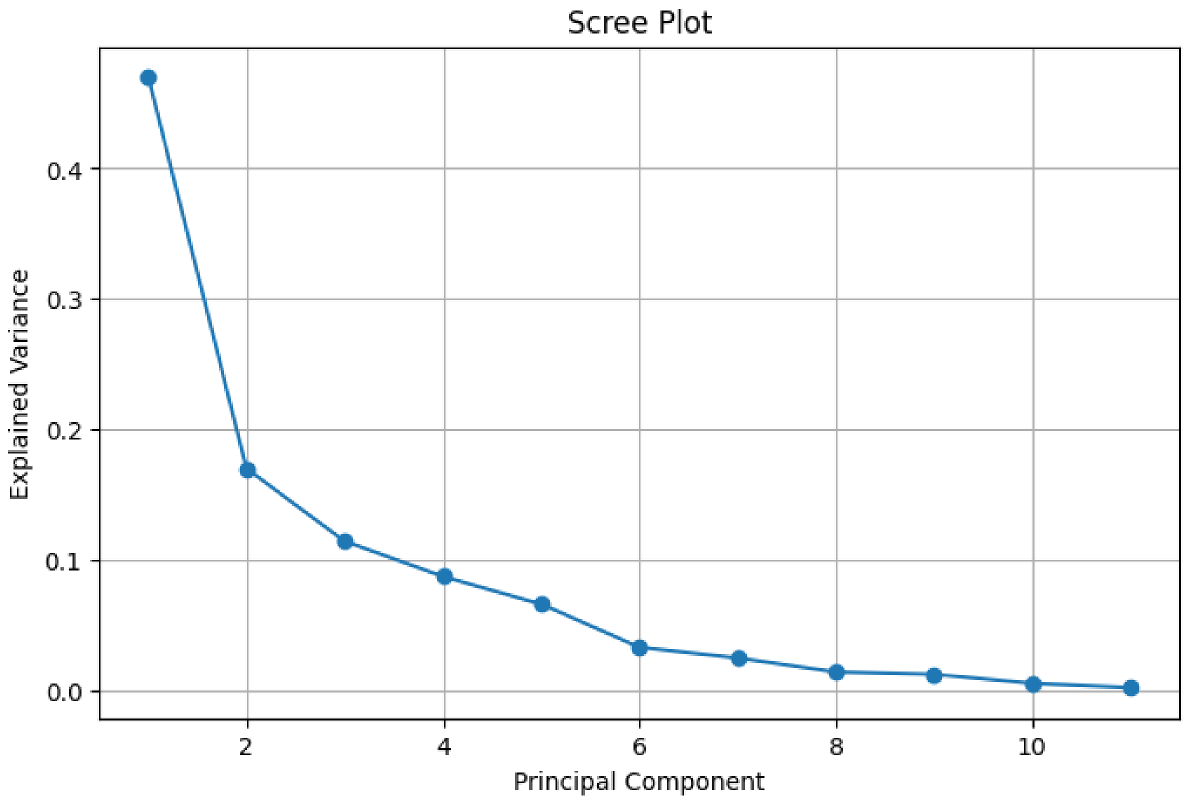

Figure 18: Scree Plot for PCA 01

A.10 PCA 02 Scree Plot (Successful Dataset)

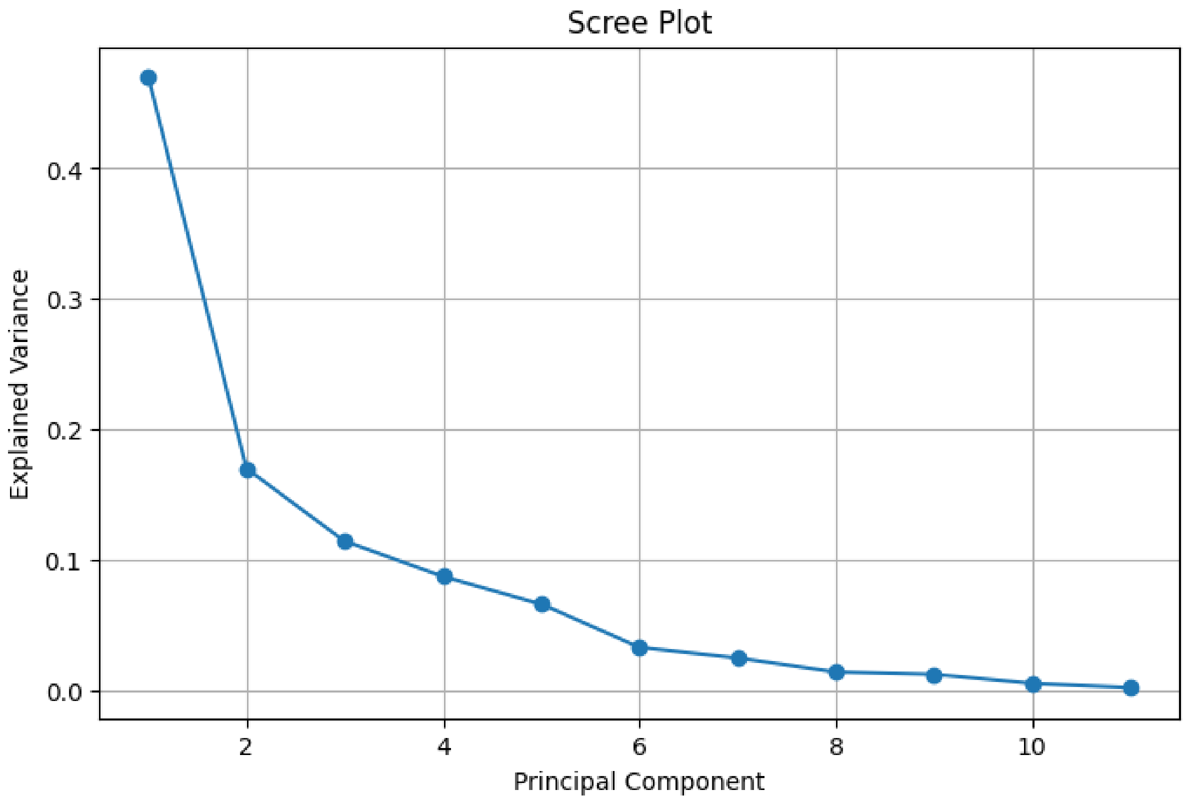

Figure 19: Scree Plot for PCA 02

A.11 K Means Clustering Elbow

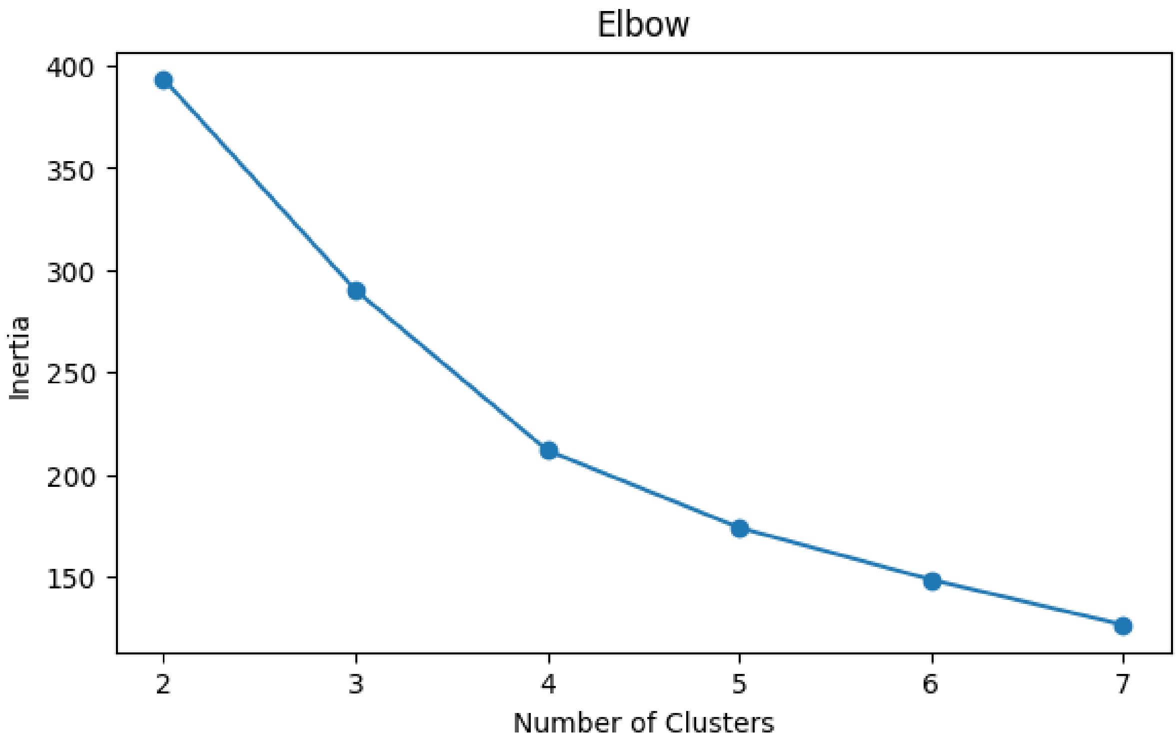

Figure 20: K Means Clustering Elbow

A.12 Signatures Per Capita by Cluster

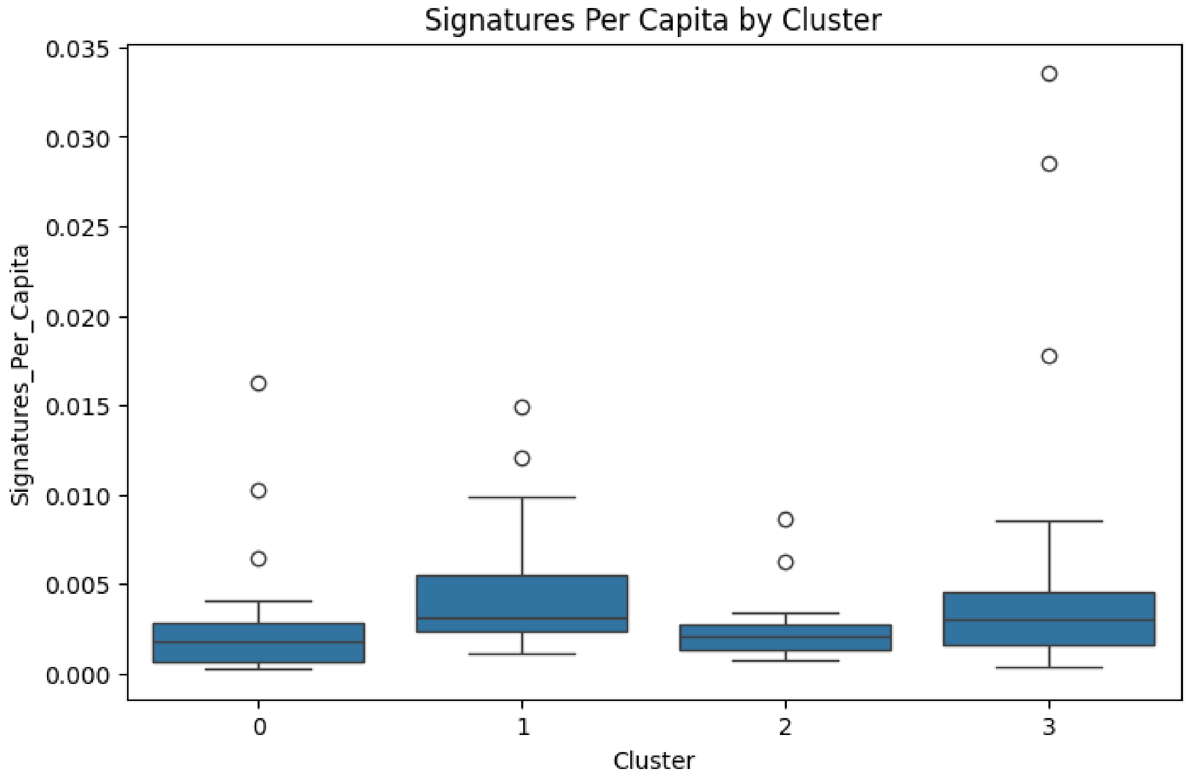

Figure 21: Signatures Per Capita by Cluster

A.13 Log Signatures Per Capita by Cluster

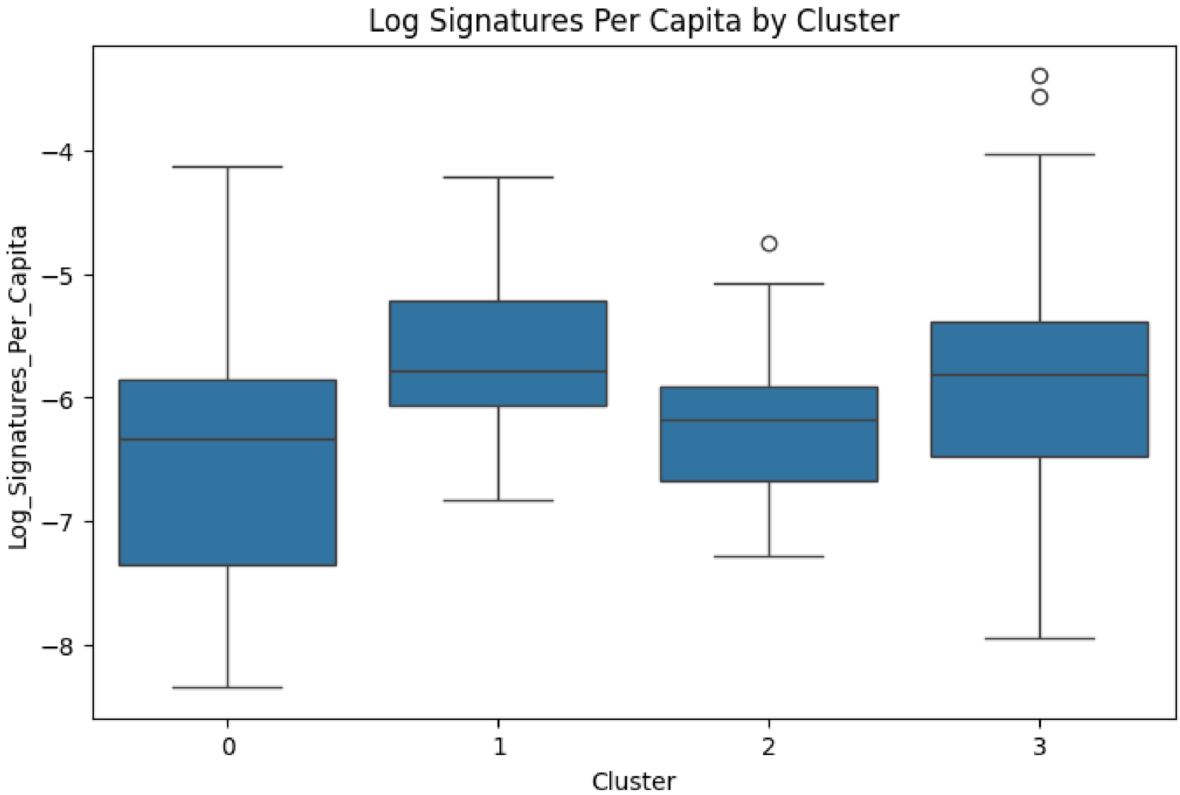

Figure 22: Log Signatures Per Capita by Cluster

A.14 Cluster Map

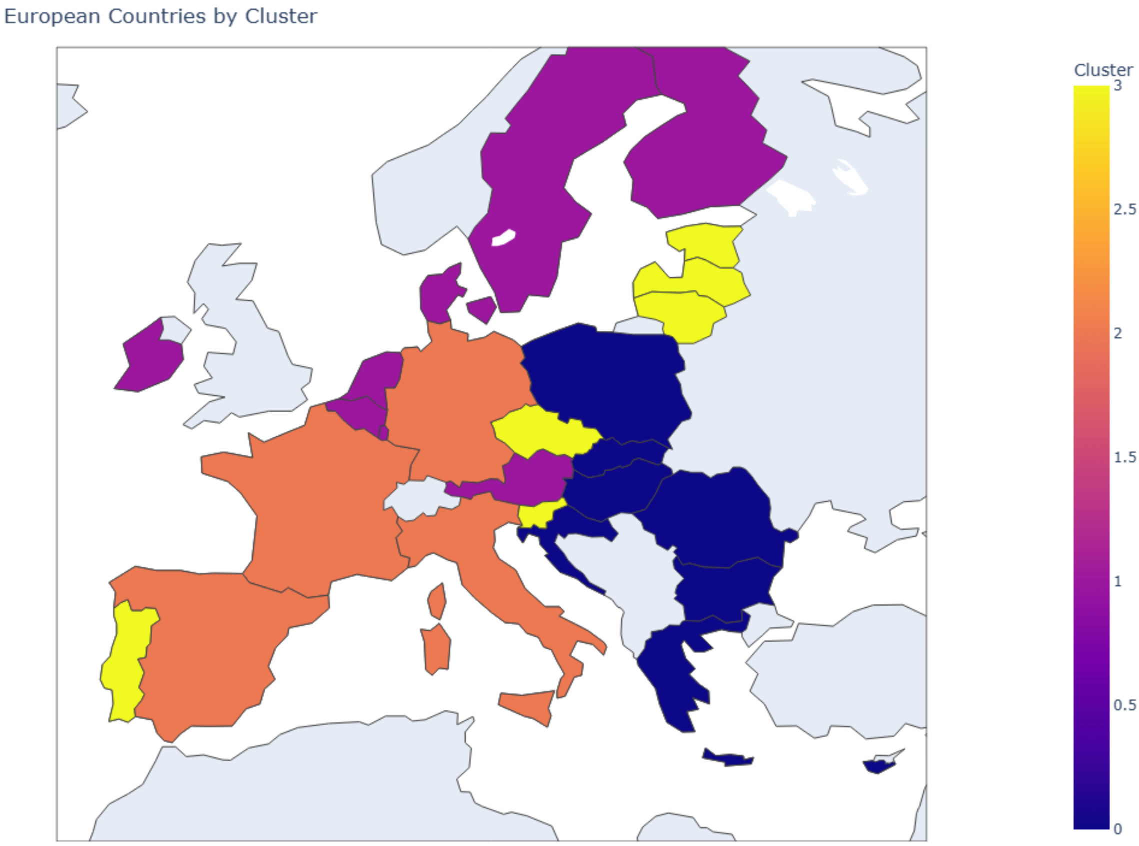

Figure 23: Country Clusters Map

A.15 Clusters vs PCA Plot

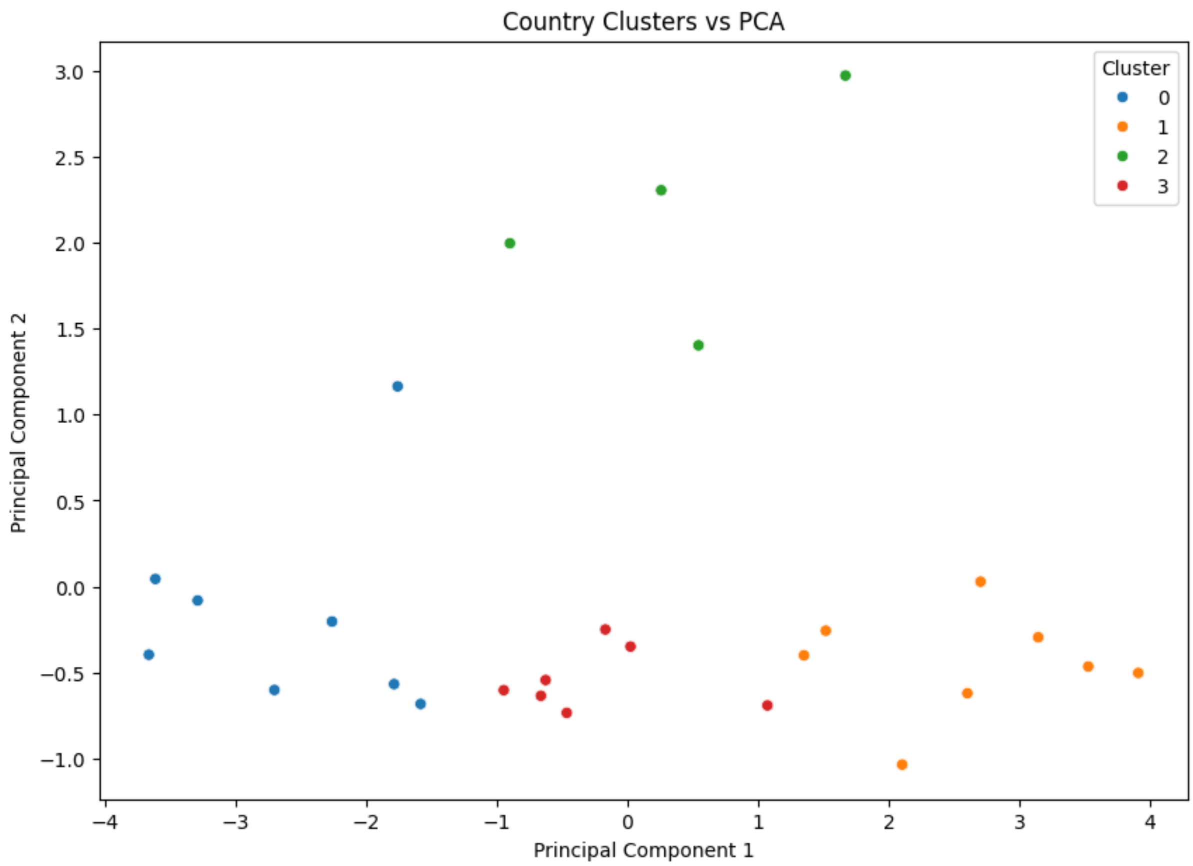

Figure 24: Clusters vs PCA Plot

A.16 Question 1 R-Squared Results

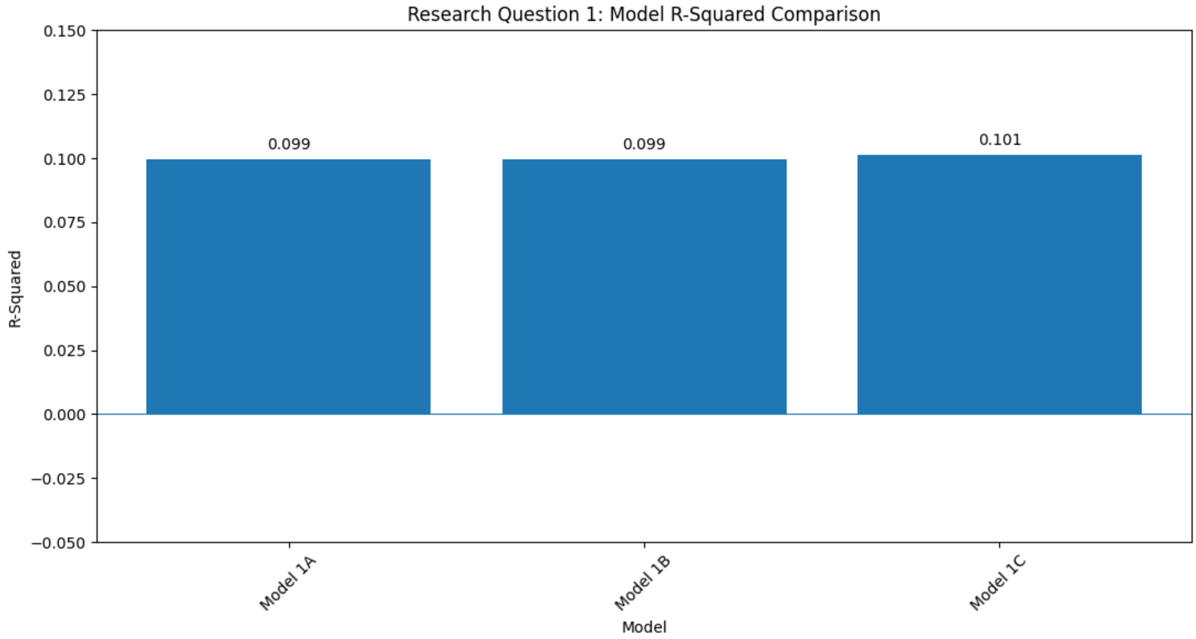

Figure 25: Question 1 R-Squared Results

A.17 Question 2 R-Squared Results

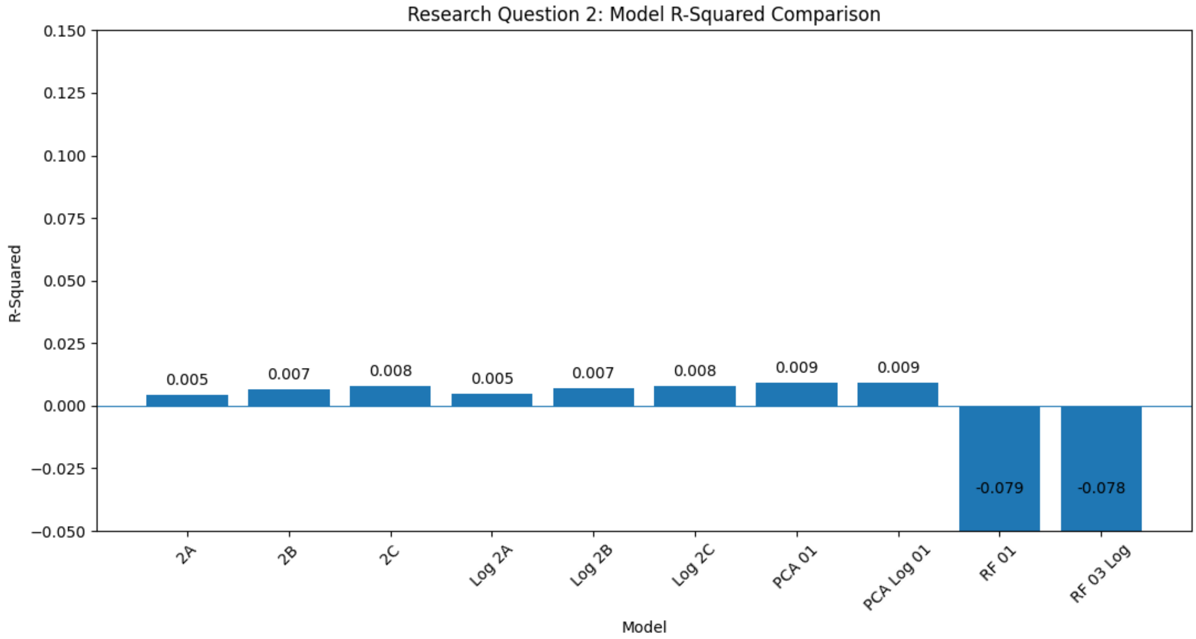

Figure 26: Question 2 R-Squared Results

A.18 Question 3 R-Squared Results

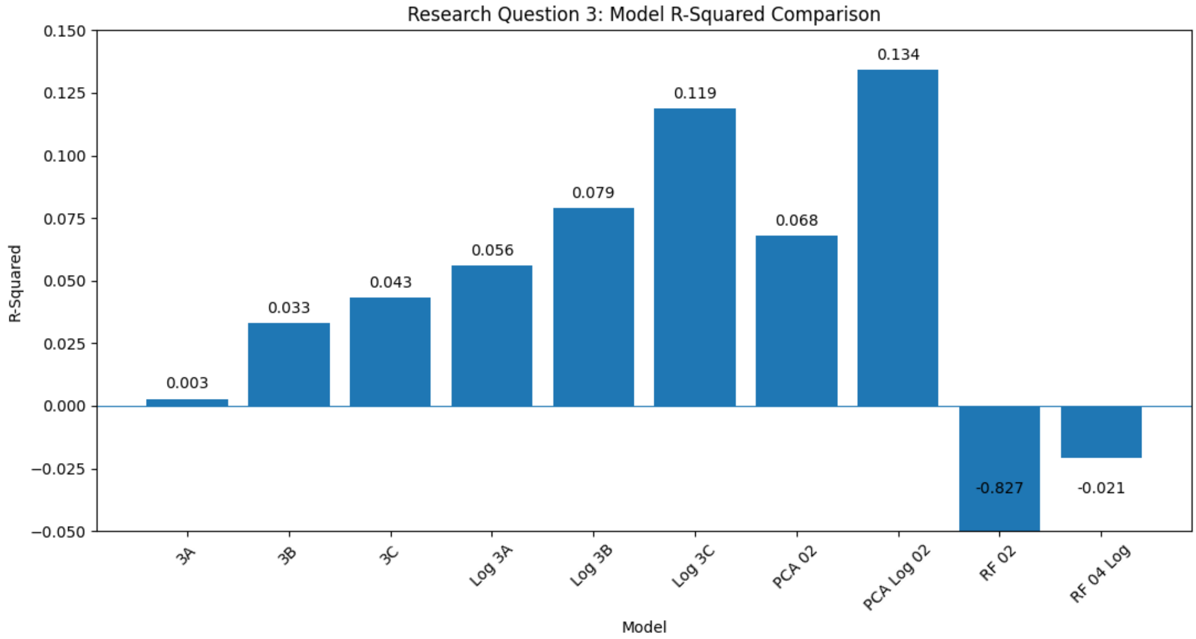

Figure 27: Question 3 R-Squared Results

A.19 Github Repository

https://github.com/Matthew-Riddell/MSc-Dissertation

A.20 Mahara Portfolio

https://mahara.dkit.ie/view/view.php?t=f816cb6da6dd833aaa2f